Resolving Tusnády's Problem

Nicola Wengiel*

Department of Mathematics, FAU Erlangen-Nürnberg

Abstract

We prove a lower bound of $\Omega_d((\log N)^{d-1/2})$ for the worst-case combinatorial discrepancy of N-point sets with respect to anchored boxes in every fixed dimension $d \geq 3$. Together with Nikolov's upper bound of the same order and Wei's planar lower bound, this settles Tusnády's problem: the discrepancy of axis-parallel boxes is $\Theta_d((\log N)^{d-1/2})$ in every fixed dimension $d \geq 2$. The proof combines smoothed product test functions, sampling discretization, and an entropy argument for conditioned Gaussian vectors.

1 Introduction

Let $P \subset \mathbb{R}^d$ be a finite set and let $\mathcal{F}$ be a family of subsets of $\mathbb{R}^d$. The *combinatorial discrepancy* of P with respect to $\mathcal{F}$ is

$$\operatorname{disc}(P, \mathcal{F}) := \min_{\chi\colon P \to \{-1,1\}} \sup_{F \in \mathcal{F}} \Big| \sum_{p \in P \cap F} \chi(p) \Big|.$$

We write $\operatorname{disc}(N, \mathcal{F})$ for its supremum over all N-point sets $P \subset \mathbb{R}^d$. Two families are of interest here: the family $\mathcal{B}_d$ of all axis-parallel boxes $\prod_{j=1}^d [a_j, b_j]$, and the family $\mathcal{A}_d$ of *anchored boxes* $A(x) := \{y \in \mathbb{R}^d : y \leq x\}$, $x \in \mathbb{R}^d$, where $y \leq x$ is understood coordinatewise. Every set $P \cap A(x)$ is of the form $P \cap B$ for a box $B \in \mathcal{B}_d$, and conversely, since P is finite, the indicator of every set $P \cap B$ with $B \in \mathcal{B}_d$ is a signed sum of indicators of 2^d sets $P \cap A(x)$ by inclusion–exclusion, so that

$$\operatorname{disc}(N, \mathcal{A}_d) \leq \operatorname{disc}(N, \mathcal{B}_d) \leq 2^d \operatorname{disc}(N, \mathcal{A}_d). \tag{1.1}$$

Tusnády's problem. In the early 1980s, Gábor Tusnády asked for the order of magnitude of $\operatorname{disc}(N, \mathcal{B}_2)$, and in particular whether it is bounded by a constant independent of N; see [Bec81], [Mat99a, p. 125] and [BN26, Section 1.7]. Beck [Bec81] answered the latter question negatively by transferring classical lower bounds for the unnormalized Lebesgue-measure discrepancy to combinatorial discrepancy for infinitely many N [Mat99a, Proposition 1.8]; for an N-point set in the unit cube, this is the largest deviation, over all boxes, of the number of points in a box from N times its volume. The smallest possible Lebesgue-measure discrepancy of an N-point set in the unit cube with respect to boxes is of order $\log N$ in the plane, the lower bound being Schmidt's theorem [Sch72], and at least of order $(\log N)^{(d-1)/2}$ in dimension d by Roth's theorem [Rot54, p. 74, footnote]. Determining the order of $\operatorname{disc}(N, \mathcal{B}_d)$ for fixed $d \geq 2$ has become known as *Tusnády's problem*.

*Email: nicola.wengiel@fau.de.

Beck also proved the upper bound $O((\log N)^4)$ in the plane; this bound and its higher-dimensional counterparts were improved in a sequence of works [Bec89, Boh90, Sri97, Mat99b, Lar14, BG17], and the currently best upper bound, due to Nikolov [Nik17, Theorem 1], is

$$\operatorname{disc}(N, \mathcal{A}_d) = O_d\big((\log N)^{d-1/2}\big) \qquad \text{for every fixed } d \geq 2. \tag{1.2}$$

On the lower bound side, Matoušek and Nikolov [MN15] and Matoušek, Nikolov and Talwar [MNT20] used the γ_2 factorization norm to prove $\operatorname{disc}(N, \mathcal{A}_d) = \Omega_d((\log N)^{d-1})$. In the plane, Wei's preprint [Wei26, Theorem 1.1] gives a lower bound of order $(\log N)^{3/2}$, matching (1.2) for $d = 2$. Thus, before the present work, the order of $\operatorname{disc}(N, \mathcal{B}_d)$ was known only up to a factor of order $\sqrt{\log N}$ in every fixed dimension $d \geq 3$. Discrepancy bounds for axis-parallel boxes have applications to numerical integration, orthogonal range searching, and differentially private range counting; see [Nik17, Section 2].

1.1 Main result and consequences

We close this gap in every fixed dimension $d \geq 3$.

Theorem 1.1. *For every integer $d \geq 3$ there are constants $c_d > 0$ and $N_d \in \mathbb{N}$ such that for every integer $N \geq N_d$,*

$$\operatorname{disc}(N, \mathcal{A}_d) \geq c_d (\log N)^{d-1/2}.$$

Corollary 1.2. *For every fixed $d \geq 3$, $\operatorname{disc}(N, \mathcal{A}_d)$ and $\operatorname{disc}(N, \mathcal{B}_d)$ are both $\Theta_d\big((\log N)^{d-1/2}\big)$.*

Proof. Combine Theorem 1.1 with (1.2) and (1.1). □

The lower bound $\Omega_d((\log N)^{d-1})$ of [MN15, MNT20] is therefore not sharp, although $(\log N)^{d-1}$ had long been conjectured to be the correct order; see Srinivasan [Sri97, p. 694], who refers to Beck and Sós [BS95], and also [Ban23, Conjecture 1.4.1] and [ABN18, Section 2.2]. Together with Wei's planar lower bound [Wei26], Corollary 1.2 settles Tusnády's problem in every fixed dimension $d \geq 2$. Since there are N-point sets in $[0,1]^d$ whose Lebesgue-measure discrepancy with respect to anchored boxes is $O_d((\log N)^{d-1})$ [Mat99a, Theorem 2.4], Theorem 1.1 also shows that for $d \geq 3$ the largest possible combinatorial discrepancy of N-point sets with respect to boxes exceeds the smallest possible Lebesgue-measure discrepancy of N-point sets by a factor of order at least $\sqrt{\log N}$, so that Beck's transference principle is not sharp in any fixed dimension $d \geq 3$.

We treat the case $d \geq 3$; for these dimensions, the relative linearization error is $O_d((\log N)^{-(d-2)})$ and tends to zero as $N \to \infty$. The constants c_d and N_d depend on d; we make no attempt to optimize them.

Theorem 1.1 extends from boxes to the homothetic copies $\mathcal{T}_B := \{tB + x : t > 0,\ x \in \mathbb{R}^d\}$ of a fixed convex polytope $B \subset \mathbb{R}^d$: the lower bound follows from it by a transfer reduction, which we carry out in Section 8.1, and the upper bound is Nikolov's [Nik17, Theorem 2]. The lower-bound constants depend only on d. Together with the lower bound of [MN15, MNT20], the transfer reduction already confirms both conjectures stated after [Nik17, Theorem 3] in every dimension $d \geq 2$, and Corollary 1.3 gives the stronger exponent $d - \frac{1}{2}$ for $d \geq 3$; see Section 8.1.

Corollary 1.3 (Fixed convex polytopes)**.** *For every fixed $d \geq 3$ and every fixed convex polytope $B \subset \mathbb{R}^d$ with nonempty interior,*

$$\operatorname{disc}(N, \mathcal{T}_B) = \Theta_{d,B}\big((\log N)^{d-1/2}\big).$$

The upper bound in Corollary 1.2 is also attained algorithmically: a randomized offline algorithm with polynomial running time colors any N-point set with discrepancy $O_d((\log N)^{d-1/2})$ for anchored boxes, with high probability (Proposition 8.2). This follows from a recent prefix-balancing theorem of Aden-Ali [AA26, Theorem 1] via the reduction in Nikolov's proof of [Nik17, Theorem 1], which we write out in Section 8.2; by Theorem 1.1, the resulting discrepancy guarantee is optimal up to the constant for $d \geq 3$. The failure probability, the number of operations and the model of computation are specified with the proof.

Notation. From Section 2 to Section 7, $d \geq 3$ is fixed; in Section 8 the range of d is stated with each result. Constants $c_d, C_d > 0$ depend only on d and may change from occurrence to occurrence, except where a statement fixes them (as c_d and N_d in Theorem 1.1); they never depend on the depth h. We write $X \lesssim Y$ or $X = O_d(Y)$ if $X \leq C_d Y$, $X = \Omega_d(Y)$ if $Y \lesssim X$, and $X \asymp Y$ or $X = \Theta_d(Y)$ if both hold. For $x \in \mathbb{R}^d$, $|x|_1 := \sum_j |x_j|$; for $p \in [0,1]^d$, $[p,1]^d := \prod_{j=1}^d [p_j, 1]$; $[n] := \{1, \dots, n\}$; and $\langle f, g \rangle := \int fg$, over the cube or over $[0,1]$. Logarithms are natural, $\mathbb{N} = \{1, 2, \dots\}$, $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$, $3\mathbb{N} = \{3, 6, \dots\}$ and $3\mathbb{N}_0 = 3\mathbb{N} \cup \{0\}$. The standard Gaussian measure on $\mathbb{R}^M$ is γ_M, and $\|\cdot\|_2$ denotes both the Euclidean norm and the norm of $L^2([0,1]^d)$. For symmetric matrices, $X \preceq Y$ means that $Y - X$ is positive semidefinite, and $\|\cdot\|_{\mathrm{op}}$ is the operator norm with respect to the Euclidean norms.

1.2 Overview of the proof

We fix $d \geq 3$ and describe the mechanism of the proof, suppressing all constants that depend only on d. The construction is parametrized by a depth $h \in 3\mathbb{N}$; the point set has $n \asymp (h+1)^{d-1} 2^h$ points, so that $h \asymp \log n$, and we show that its *hereditary* discrepancy for anchored boxes, that is, the largest discrepancy of one of its subsets, is at least $c_d(h+1)^{d-1/2}$ (Proposition 7.2). Hence some subset has discrepancy at least $c_d(h+1)^{d-1/2}$, and padding this subset with points outside all relevant anchored boxes yields Theorem 1.1 for every large N.

Duality. For points $P_1, \dots, P_n \in (0,1)^d$ and a weight vector $\eta \in \mathbb{R}^n$, let $G_\eta(x) := \sum_{i \leq n} \eta_i \mathbf{1}_{\{P_i \leq x\}}$, $x \in [0,1]^d$, be the weighted count of the anchored box $A(x)$. A coloring is a weight vector $\eta \in \{-1, 1\}^n$, and its discrepancy is $\|G_\eta\|_\infty := \sup_{x \in [0,1]^d} |G_\eta(x)|$. For an integrable function W on the cube, Fubini's theorem and Hölder's inequality give

$$\langle G_\eta, W \rangle = \sum_{i=1}^n \eta_i \, (J^* W)(P_i) \leq \|G_\eta\|_\infty \, \|W\|_1, \qquad (J^* W)(p) := \int_{[p,1]^d} W(x) \, dx. \tag{1.3}$$

A test function W with $\|W\|_1 \leq 1$ and a large value of $\sum_i \eta_i (J^* W)(P_i)$ thus certifies a large weighted box count. The difficulty is that the coloring is chosen adversarially after the point set, so that the test function must be adapted to it; we obtain this adaptation from a probabilistic reformulation in which the adversarial coloring is replaced by a Gaussian weight vector conditioned on a small-ball event. Concretely, we construct a family of test functions $W(u)$, $u \in [-1,1]^M$, and evaluate it at $u = U(Z)$, the Rosenblatt coordinates of the conditioned Gaussian vector Z: if the hereditary discrepancy is small, the conditioning event has probability at least $\mathrm{e}^{-O(M)}$ (Step 1); this forces Z to correlate with $U(Z)$ (Step 2); and the test functions are designed so that this correlation produces a large weighted box count on average, contradicting the conditioning (Steps 3–5).

Step 1: hereditary discrepancy and Gaussian small balls. Let D be the hereditary discrepancy of the point set for anchored boxes, and let η be a centered Gaussian vector in $\mathbb{R}^n$ with all coordinate variances at most one. Lemma 7.1 shows that

$$\mathbb{P}(\|G_\eta\|_\infty \le 2D) \ge (p_0/2)^n, \qquad p_0 := \mathbb{P}(|g| \le 1), \tag{1.4}$$

where g is a standard Gaussian variable, by combining Šidák's inequality [Š67], the rounding lemma of Lovász, Spencer and Vesztergombi [LSV86], and Anderson's inequality [And55].

Step 2: conditioned Gaussian vectors and Rosenblatt coordinates. We take $\eta = Bz$, where z is a standard Gaussian vector in $\mathbb{R}^M$, M is the number of shell boxes introduced in Step 3, and the matrix $B\colon \mathbb{R}^M \to \mathbb{R}^n$, constructed in Step 4, satisfies $BB^* \preceq I_n$. Let $K_t := \{z : \|G_{Bz}\|_\infty \le t\}$, a closed, convex and symmetric set (from Section 5 on we abbreviate $G_z := G_{Bz}$). Since $\eta = Bz$ has coordinate variances $(BB^*)_{ii} \le 1$ and $n \le C_d M$, (1.4) gives $\gamma_M(K_{2D}) \ge \mathrm{e}^{-\kappa M}$ for a constant κ depending only on d.

Whenever $\gamma_M(K_t) \ge \mathrm{e}^{-\kappa M}$, the conditioned vector Z with law $\nu := \gamma_M(\cdot \mid K_t)$ has differential entropy at least $\frac{M}{2}\log(2\pi) - \kappa M$, which is at most $\kappa + \frac{1}{2}$ nats per coordinate below the entropy of γ_M. Its Rosenblatt transform $U = (U_1, \dots, U_M)$, $U_j := 2F_j(Z_j \mid Z_{<j}) - 1$ [Ros52], built from the successive conditional distribution functions F_j, has independent coordinates that are uniform on $[-1, 1]$, and an elementary inequality between entropy and the Gini mean difference, combined with the chain rule for entropy, yields

$$\sum_{j=1}^{M} \mathbb{E}_\nu(Z_j U_j) \ge c_\kappa M \tag{1.5}$$

with $c_\kappa > 0$ depending only on κ (Proposition 6.2). Each $\mathbb{E}_\nu(Z_j U_j)$ is nonnegative, and their average is bounded below by c_κ. The lower bound $\gamma_M(K_t) \ge \mathrm{e}^{-\kappa M}$ enters the argument only through (1.5).

Step 3: test functions with linear response. The heart of the construction is a family of functions $W(u)$ on $[0,1]^d$, indexed by coefficient vectors $u \in [-1,1]^M$, with $\|W(u)\|_1 \le 1$ for every u and such that $J^* P_{\mathrm{hi}} W(u)$, the primitive of $W(u)$ after removal of its component in the *low Haar cross*, the span of all tensor Haar functions of level below h, is linear in u up to an error that is small in quadratic mean over u; see (1.7). The index set is a family $\mathcal{R}$ of M dyadic boxes of volume 2^{-h} whose sides have depths in $3\mathbb{N}_0$ (hence $h \in 3\mathbb{N}$), the *shell boxes*, to each of which a localized Walsh function q_R is attached; from now on we index the coordinates of $\mathbb{R}^M$ by $\mathcal{R}$. The test function is the normalized difference

$$W(u) := \tfrac{1}{2}\big(Q_+(u) - Q_-(u)\big), \qquad Q_\pm(u) := \prod_a \big(1 \pm \varepsilon B_a(u)\big), \qquad \varepsilon = \tfrac{1}{4}, \tag{1.6}$$

of two positive products. Here the shapes of the shell boxes are grouped into blocks by the depth a of their first side, and the block function $B_a(u)$ is obtained by inserting the layer functions $\sum_R u_R q_R$ (one for each shape of the block, R running over the shell boxes of that shape) into the multilinear extension of $\xi \mapsto \tanh(\theta \sum_r \xi_r)$ from the Boolean cube (Section 3). Because every B_a is odd in one binary digit of the first coordinate that no block of smaller first depth sees, both products integrate to one along every line parallel to the first axis, which gives $\|W(u)\|_1 \le 1$ (Lemma 3.2).

The linear part of $W(u)$ in u is $W_{\mathrm{lin}}(u) = \sum_R \lambda_R u_R q_R$ with $\lambda_R \asymp \theta$, where the modulation parameter is $\theta := t_d (h+1)^{-(d-2)/2}$ with a small constant t_d. The nonlinear part is not small as a function, but

it becomes small after removal of its component in the low Haar cross and integration: Theorem 4.1 shows that

$$\mathbb{E}_U\big(\|J^*P_{\mathrm{hi}}W(U) - J^*W_{\mathrm{lin}}(U)\|_2^2\big) \le C_d\theta^4\ \mathbb{E}_U\big(\|J^*W_{\mathrm{lin}}(U)\|_2^2\big) \tag{1.7}$$

for uniform random coefficients U, where P_{hi} denotes the orthogonal projection onto the orthogonal complement of the low Haar cross. The relative root-mean-square error is thus $O_d(\theta^2) = O_d((h+1)^{-(d-2)})$, which tends to zero as $h \to \infty$ because $d \ge 3$.

Step 4: sampling discretization. By (1.3), $\langle G_{Bz}, W(u)\rangle = z \cdot V(u)$ with the *response* $V(u) := B^*\big((J^*W(u))(P_i)\big)_{i\le n}$. The points come from a sampling discretization theorem of Limonova and Temlyakov [LT22, Theorem 1.1], applied to a space of piecewise multiaffine functions that contains the primitives J^*q_R and the primitives of the low Haar cross, and the matrix B is built from the quadrature weights, the evaluations of the primitives, a projection that annihilates the low Haar cross, and an inverse Gram matrix. Since the remainder $J^*P_{\mathrm{hi}}W(u) - J^*W_{\mathrm{lin}}(u)$ need not lie in this space, the sampling space is enlarged by $\sqrt{w}$, where $w(p) := \mathbb{E}_U(|(J^*P_{\mathrm{hi}}W(U) - J^*W_{\mathrm{lin}}(U))(p)|^2)$ is the mean-square remainder; the quadrature rule then also bounds the weighted sum of the values $w(P_i)$ by a constant multiple of $\int w$, which transfers (1.7) to the points (Lemma 5.3).

Proposition 5.6 shows that the response is diagonal in u up to a small error: there are numbers $s_R > 0$, all comparable to a common scale $s \asymp \theta\,(h+1)^{(d-1)/2}2^{-h}$, which results from our chosen normalization of B, such that

$$V(u) = \mathrm{diag}(s_R)\,u + e(u), \qquad s_* := \min_R s_R \asymp s, \qquad \mathbb{E}_U\big(\|e(U)\|_2^2\big) \le \rho_h^2 s_*^2 M, \tag{1.8}$$

with $\rho_h := C_\rho(h+1)^{-(d-2)}$ for a constant C_ρ depending only on d.

Step 5: conclusion. For $z \in K_t$ and every $u \in [-1,1]^M$ we have $z \cdot V(u) = \langle G_{Bz}, W(u)\rangle \le t$. Substituting $u = U(z)$, where U is the Rosenblatt transform of ν, taking expectations under ν, and using (1.8), (1.5) together with $\mathbb{E}_\nu(Z_RU_R) \ge 0$ for every R, the exact uniform law of U under ν, and $\mathbb{E}_\nu(\|Z\|_2^2) \le M$ (Lemma 2.4), we obtain

$$t \ \ge\ \mathbb{E}_\nu(Z \cdot V(U)) \ \ge\ s_* \sum_R \mathbb{E}_\nu(Z_RU_R) - \sqrt{M}\,\big(\mathbb{E}_U\big(\|e(U)\|_2^2\big)\big)^{1/2} \ \ge\ (c_\kappa - \rho_h)\,s_*M \tag{1.9}$$

(Proposition 6.3). With $t = 2D$ and h so large that $\rho_h \le c_\kappa/2$, this gives $D \gtrsim s_*M$. The exponent $d - 1/2$ arises as follows: the number $M \asymp (h+1)^{d-1}2^h$ of shell boxes and the factor $s/\theta \asymp (h+1)^{(d-1)/2}2^{-h}$ of the response scale give $s_*M \asymp \theta\,(h+1)^{3(d-1)/2}$, and the modulation parameter $\theta \asymp (h+1)^{-(d-2)/2}$ reduces this to $(h+1)^{d-1/2} \asymp (\log n)^{d-1/2}$.

Relation to earlier methods. Products of modulated block sums such as $Q_\pm$ in (1.6) go back to Riesz products, which appear in Halász's lower bounds for the Lebesgue-measure discrepancy [Hal81] and in proofs of small-ball inequalities: in Temlyakov's Riesz-product proof [Tem95b] of Talagrand's two-dimensional Haar inequality [Tal94, Proposition 3.2] and its trigonometric analogue [Tem95a], and in the higher-dimensional results of Bilyk and Lacey [BL08] and Bilyk, Lacey and Vagharshakyan [BLV08], where coincidences between Haar functions of equal depth in the same coordinate are the main obstacle. Here the product is paired with the counting function of a Gaussian weight vector, its coefficients are the Rosenblatt coordinates of that vector, and the coincidences are controlled: since the depths of the shell boxes are spaced by three, every nonvanishing product of the functions q_I in one coordinate reduces to a factor $\pm q_{I'}$ or $\pm\mathbf{1}_{I'}$, whose contributions are controlled by the smoothing

of the block functions (Lemma 2.3) and by the estimate of the high-frequency primitives (Section 4). The passage from colorings to Gaussian vectors through hereditary discrepancy is in the spirit of the factorization-norm approach of [MNT20], but the quantity we estimate is a Gaussian small-ball probability rather than a γ_2 norm: hereditary discrepancy bounds it from below (Lemma 7.1), and the entropy argument bounds it from above for radii $t \ll (h+1)^{d-1/2}$ (Proposition 6.3).

After the preliminaries in Section 2, Sections 3–6 develop the main construction and entropy argument, Section 7 completes the proof of Theorem 1.1, and Section 8 establishes the extensions and algorithmic consequences.

2 Preliminaries

This section collects definitions, external results and auxiliary lemmas. Section 2.1 and the Gaussian inequalities of Section 2.4 are used in Step 1 of the overview (Section 7); the Rosenblatt transform and the entropy facts of Sections 2.4 and 2.5 in Step 2 and the second-moment bound of Lemma 2.4 in Step 5 (both in Section 6); the dyadic functions and the Walsh estimate of Sections 2.2 and 2.3 in the construction of the test functions (Sections 3 and 4), the dyadic functions also in Section 5; and the sampling theorem of Section 2.6 in Section 5. Lemmas 2.2 and 2.3 are specific to our construction; the remaining lemmas are standard and included for completeness.

2.1 Discrepancy and hereditary discrepancy

For a real matrix A with n columns, its discrepancy and hereditary discrepancy are

$$\operatorname{disc}(A) := \min_{\sigma\in\{-1,1\}^n} \|A\sigma\|_\infty, \qquad \operatorname{herdisc}(A) := \max_{S\subseteq[n]} \operatorname{disc}(A_S),$$

where A_S is the submatrix of A formed by the columns indexed by S. For a finite point set $P = \{P_1, \dots, P_n\} \subset \mathbb{R}^d$, the set system of anchored boxes with corners in the unit cube is $\{P \cap A(x) : x \in [0,1]^d\}$; its incidence matrix $A_P \in \{0,1\}^{m\times n}$ has one row $(\mathbf{1}_{\{P_i \le x\}})_{i\le n}$ for each of the finitely many distinct sets $P \cap A(x)$. For every subset $S \subseteq P$, the columns of A_P indexed by S form, up to repeated rows, the incidence matrix of the anchored boxes on S, so that

$$\operatorname{herdisc}(A_P) = \max_{S\subseteq P} \min_{\chi\colon S\to\{-1,1\}} \sup_{x\in[0,1]^d} \Big| \sum_{p\in S\cap A(x)} \chi(p) \Big| \le \max_{S\subseteq P} \operatorname{disc}(S, \mathcal{A}_d). \tag{2.1}$$

The inequality holds because $\mathcal{A}_d$ also contains anchored boxes with corners outside the unit cube; (2.1) is used in the proof of Theorem 1.1. We shall use the inequality between linear and hereditary discrepancy due to Lovász, Spencer and Vesztergombi [LSV86, Corollary 1] (their discrepancy is defined with colorings in $\{0,1\}$ and is half of ours, so that their Corollary 1 is the first assertion of Lemma 2.1); see also [Mat99a, Theorem 4.6]. We state it in the form (2.2), which is all we need, and include the short proof by dyadic rounding for completeness.

Lemma 2.1 (Rounding lemma)**.** *Let $A \in \mathbb{R}^{m\times n}$ satisfy $\operatorname{herdisc}(A) \le D$. Then for every $y \in [0,1]^n$ there is $b \in \{0,1\}^n$ with $\|A(y-b)\|_\infty \le D$. Consequently,*

$$A[-1,1]^n \subseteq \bigcup_{\sigma\in\{-1,1\}^n} \big(A\sigma + [-2D, 2D]^m\big). \tag{2.2}$$

Proof. Let y first have dyadic coordinates with common denominator 2^k, and iterate for $j = k, k-1, \dots, 1$: let S be the set of coordinates whose numerator (in denominator 2^j) is odd, choose

$\sigma \in \{-1,1\}^S$ with $\|A_S\sigma\|_\infty \le D$, and add $2^{-j}\sigma$ to the coordinates in S. Odd numerators o satisfy $0 < o < 2^j$, so all coordinates stay in $[0,1]$, all numerators become even, and Ay changes by at most $2^{-j}D$ in the maximum norm. After the step $j = 1$ the vector $b \in \{0,1\}^n$ satisfies $\|A(y-b)\|_\infty \le D\sum_{j=1}^{k} 2^{-j} \le D$. For general y, approximate by dyadic vectors $y^{(\ell)} \to y$, pass to a subsequence along which the resulting $b^{(\ell)} = b$ is constant, and let $\ell \to \infty$. For (2.2), apply this to $y := (\eta + \mathbf{1})/2$ with $\eta \in [-1,1]^n$ and put $\sigma := 2b - \mathbf{1}$; then $A\eta - A\sigma = 2A(y-b)$. □

2.2 Dyadic intervals, Rademacher and Haar functions, primitives

For $j \in \mathbb{N}_0$, let $\mathcal{D}_j$ be the set of the 2^j half-open dyadic intervals $[k2^{-j}, (k+1)2^{-j})$, $0 \le k < 2^j$, of *depth* j; we write $\mathrm{depth}(I) := j$ for $I \in \mathcal{D}_j$, and $|I| = 2^{-j}$ for its length. Two dyadic intervals are either nested or disjoint. For $I \in \mathcal{D}_j$ and $0 \le b \le j$, the *ancestor* of I of depth b is the unique interval of $\mathcal{D}_b$ containing I. The *Rademacher functions* are $\varrho_j(x) := (-1)^{\lfloor 2^j x \rfloor}$ for $x \in [0,1)$ and $j \ge 1$; ϱ_j is constant on every interval of $\mathcal{D}_j$ and takes the values $+1$ and -1 on the left and right halves of every interval of $\mathcal{D}_{j-1}$. The (L^∞-normalized) *Haar function* of $J \in \mathcal{D}_b$ is

$$h_J := \varrho_{b+1}\mathbf{1}_J = \mathbf{1}_{J^-} - \mathbf{1}_{J^+},$$

where J^- and J^+ are the left and right halves of J. Its *level* is $\mathrm{lev}(h_J) := b$, and the constant function $\mathbf{1}$ has level $\mathrm{lev}(\mathbf{1}) := -1$. The functions $\mathbf{1}$ and h_J, $J \in \bigcup_b \mathcal{D}_b$, form an orthogonal basis of $L^2[0,1]$, and the *tensor Haar functions*

$$\varphi_1 \otimes \cdots \otimes \varphi_d, \qquad \varphi_j \in \{\mathbf{1}\} \cup \{h_J : J \in \textstyle\bigcup_b \mathcal{D}_b\},$$

form an orthogonal basis of $L^2([0,1]^d)$. The level of a tensor Haar function is the sum $\sum_j \mathrm{lev}(\varphi_j) \in \{-d, -d+1, \dots\}$ of its coordinate levels; for every integer L there are only finitely many tensor Haar functions of level at most L.

Recall from (1.3) the *primitive*

$$(J^*f)(p) := \int_{[p,1]^d} f(x)\,dx, \qquad p \in [0,1]^d, \tag{2.3}$$

of an integrable function f on $[0,1]^d$. The function J^*f is continuous, and J^* is linear and acts on tensor products coordinatewise: $J^*(f_1 \otimes \cdots \otimes f_d) = J^*f_1 \otimes \cdots \otimes J^*f_d$, where in one dimension $(J^*g)(t) = \int_t^1 g$ for $t \in [0,1]$. With the variable $t \in [0,1]$, one computes

$$J^*\mathbf{1} = 1 - t, \quad \|J^*\mathbf{1}\|_2^2 = \tfrac{1}{3}, \qquad J^*h_J = -\tau_J, \quad \|J^*h_J\|_2^2 = \tfrac{|J|^3}{12}, \tag{2.4}$$

where τ_J is the tent function supported on J that is affine on J^- and on J^+ and has height $|J|/2$ at the midpoint of J.

Localized Walsh functions q_I and their primitives ψ_I. The functions q_I are the one-dimensional factors of the localized Walsh functions q_R attached to the shell boxes in Section 3. The three Rademacher factors make q_I orthogonal to all polynomials of degree at most two, so that ψ_I is orthogonal to affine functions (Lemma 2.2); this orthogonality is used in Sections 3 and 5. For $I \in \mathcal{D}_a$ we define

$$q_I := \mathbf{1}_I\, \varrho_{a+1}\varrho_{a+2}\varrho_{a+3} = h_{I_0} - h_{I_1} - h_{I_2} + h_{I_3}, \qquad \psi_I := J^*q_I, \tag{2.5}$$

where $I_0, I_1, I_2, I_3 \in \mathcal{D}_{a+2}$ are the four quarters of I from left to right. The second expression follows from the sign pattern of $\varrho_{a+1}\varrho_{a+2}$ on the quarters and from $\varrho_{a+3}\mathbf{1}_{I_k} = h_{I_k}$. Thus $|q_I| = \mathbf{1}_I$, q_I is a combination of Haar functions of level $a+2$, and q_I is constant on every interval of $\mathcal{D}_{a+3}$.

Lemma 2.2 (Moments of q_I and ψ_I)**.** *Let $I \in \mathcal{D}_a$. Then $\int q_I(x)x^k\,dx = 0$ for $k = 0, 1, 2$. The primitive ψ_I is continuous, vanishes outside I, is affine on every interval of $\mathcal{D}_{a+3}$, and satisfies*

$$\int_0^1 \psi_I(p)\,dp = \int_0^1 p\,\psi_I(p)\,dp = 0, \qquad \|\psi_I\|_2^2 = \frac{|I|^3}{192}. \tag{2.6}$$

In particular, $\langle \psi_I, g\rangle = 0$ for every function g that is affine on an interval containing I.

Proof. Write $I = [c, c+\ell)$ and substitute $x = c + \ell u$ with $u \in [0,1)$; then $\varrho_{a+k}(x) = \varrho_k(u)$ for $k = 1, 2, 3$, so the moments are $\ell \int_0^1 \varrho_1\varrho_2\varrho_3(u)\,(c+\ell u)^k\,du$. Writing $u = \sum_{i\ge1} \iota_i 2^{-i}$ with binary digits $\iota_i = (1 - \varrho_i(u))/2$, the function u^k with $k \le 2$ is an absolutely convergent series of products of at most two Rademacher functions, while $\varrho_1\varrho_2\varrho_3$ is a product of three. Since the products $\prod_{i\in T}\varrho_i$ over distinct finite sets $T \subset \mathbb{N}$ are orthonormal in $L^2[0,1]$, integrating termwise shows that the moments vanish for $k \le 2$. Since q_I vanishes outside I and $\int q_I = 0$, the primitive $\psi_I(p) = \int_p^1 q_I$ vanishes for $p \notin I$. Fubini's theorem gives $\int_0^1 \psi_I(p)\,dp = \int q_I(x)x\,dx = 0$ and $\int_0^1 p\,\psi_I(p)\,dp = \frac12\int q_I(x)x^2\,dx = 0$; as ψ_I vanishes outside I, the assertion on affine functions g follows. On I, the function q_I equals ±1 on the eight intervals of $\mathcal{D}_{a+3}$ contained in I, with the sign pattern $(+,-,-,+,-,+,+,-)$; integrating from the right, ψ_I is the sum of four tents of height $\ell/8$ and base width $\ell/4$ with signs $(-,+,+,-)$, which gives $\|\psi_I\|_2^2 = 4\cdot\frac{\ell}{4}\cdot\frac{(\ell/8)^2}{3} = \frac{\ell^3}{192}$. The remaining assertions (continuity, affinity on the intervals of $\mathcal{D}_{a+3}$) follow from this representation as a sum of tents. □

2.3 Fourier–Walsh expansions

Every function $F\colon \{-1,1\}^m \to \mathbb{R}$ has the expansion $F = \sum_{T\subseteq[m]} \widehat{F}(T)\chi_T$ with $\chi_T(\xi) := \prod_{i\in T}\xi_i$ and $\widehat{F}(T) = \mathbb{E}(F(\xi)\chi_T(\xi))$ for a uniformly distributed ξ; see [O'D14, Chapter 1 and Section 2.2]. The characters χ_T are orthonormal, so $\mathbb{E}(F(\xi)^2) = \sum_T \widehat{F}(T)^2$. The *multilinear extension* of F is

$$\widetilde{F}(y) := \sum_{T\subseteq[m]} \widehat{F}(T)\prod_{i\in T} y_i, \qquad y \in [-1,1]^m. \tag{2.7}$$

If $\xi_1, \dots, \xi_m$ are independent random signs with $\mathbb{E}(\xi_i) = y_i$, then $\widetilde{F}(y) = \mathbb{E}(F(\xi))$; in particular $|\widetilde{F}(y)| \le \max|F|$. For $i \in [m]$, the discrete derivative $D_iF(\xi) := \frac12\big(F(\xi^{i\to+}) - F(\xi^{i\to-})\big)$, where $\xi^{i\to\pm}$ agrees with ξ except that its i-th coordinate is ±1, has the expansion $D_iF = \sum_{T\ni i}\widehat{F}(T)\chi_{T\setminus\{i\}}$ [O'D14, Proposition 2.19]. The operators D_i commute, and for distinct i, j, k,

$$\mathbb{E}\big((D_iF)^2\big) = \sum_{T\ni i}\widehat{F}(T)^2, \qquad \mathbb{E}\big((D_iD_jD_kF)^2\big) = \sum_{T\supseteq\{i,j,k\}}\widehat{F}(T)^2. \tag{2.8}$$

The block functions of Section 3 are multilinear extensions of the function in the following lemma. Part (ii) identifies their linear part, and part (iii) controls the nonlinear part in Section 4.

Lemma 2.3 (Walsh coefficients of a smoothed block sum)**.** *Let $m \in \mathbb{N}$, $\theta > 0$, and $F(\xi) := \tanh\big(\theta\sum_{i=1}^m \xi_i\big)$ for $\xi \in \{-1,1\}^m$. Then:*

(i) *$\widehat{F}(T) = 0$ whenever $|T|$ is even;*

(ii) *the first-order coefficient $b := \widehat{F}(\{i\})$ does not depend on i, and $\theta\big(1 - \theta^2(m - \frac23)\big) \le b \le \theta$;*

(iii) *for every $\mathcal{E} \subseteq [m]$,*

$$\sum_{\emptyset\ne T\subseteq\mathcal{E}}\widehat{F}(T)^2 \le \theta^2|\mathcal{E}| \qquad \text{and} \qquad \sum_{T\subseteq\mathcal{E},\ |T|\ge3}\widehat{F}(T)^2 \le \theta^6|\mathcal{E}|^3. \tag{2.9}$$

Proof. (i) F is odd, $F(-\xi) = -F(\xi)$, and $\chi_T(-\xi) = (-1)^{|T|}\chi_T(\xi)$, so $\widehat{F}(T) = -(-1)^{|T|}\widehat{F}(T)$.

(ii) By symmetry b does not depend on i. Let $S := \sum_{l\neq i}\xi_l$ and $g(s) := \tanh(\theta s)$. Then $b = \mathbb{E}(\xi_i F(\xi)) = \frac{1}{2}\,\mathbb{E}(g(S+1) - g(S-1)) = \frac{1}{2}\,\mathbb{E}(\int_{-\theta}^{\theta}\operatorname{sech}^2(\theta S + t)\,dt)$. Since $\operatorname{sech}^2 \le 1$, $b \le \theta$. Since $\operatorname{sech}^2(y) \ge 1 - y^2$ and $\mathbb{E}((\theta S + t)^2) = \theta^2(m-1) + t^2$,

$$b \ge \frac{1}{2}\int_{-\theta}^{\theta}\big(1 - \theta^2(m-1) - t^2\big)dt = \theta - \theta^3(m-1) - \frac{\theta^3}{3}.$$

(iii) Since tanh is 1-Lipschitz, $|D_i F| \le \theta$, and (2.8) gives $\sum_{\emptyset\neq T\subseteq\mathcal{E}}\widehat{F}(T)^2 \le \sum_{i\in\mathcal{E}}\sum_{T\ni i}\widehat{F}(T)^2 \le \theta^2|\mathcal{E}|$. For distinct i,j,k, with g as in (ii) and $S' := \sum_{l\notin\{i,j,k\}}\xi_l$,

$$D_iD_jD_kF(\xi) = \frac{1}{8}\big(g(S'+3) - 3g(S'+1) + 3g(S'-1) - g(S'-3)\big) = g'''(\zeta)$$

for some $\zeta \in (S'-3, S'+3)$, by the mean value theorem for third-order differences with step two. As $|\tanh'''| \le 2$, we get $|D_iD_jD_kF| \le 2\theta^3$. Every $T \subseteq \mathcal{E}$ with $|T| \ge 3$ contains a triple $\{i,j,k\} \subseteq \mathcal{E}$, so by (2.8)

$$\sum_{T\subseteq\mathcal{E},\ |T|\ge 3}\widehat{F}(T)^2 \le \sum_{\{i,j,k\}\subseteq\mathcal{E}}\mathbb{E}\big((D_iD_jD_kF)^2\big) \le \binom{|\mathcal{E}|}{3}4\theta^6 \le \theta^6|\mathcal{E}|^3. \qquad \square$$

2.4 Gaussian inequalities and the Rosenblatt transform

We call a random vector η in $\mathbb{R}^n$ a *centered Gaussian vector* if $\eta = Bz$ for some $k \in \mathbb{N}$, a standard Gaussian vector z in $\mathbb{R}^k$ and a matrix $B \in \mathbb{R}^{n\times k}$; its covariance BB^* (B^* the transpose) may be singular. We use two classical inequalities, a second-moment bound and the Rosenblatt transform.

Theorem A (Šidák [Š67, Corollary 1]). *Let η be a centered Gaussian vector in $\mathbb{R}^n$ and $c_1, \dots, c_n > 0$. Then*

$$\mathbb{P}(|\eta_i| \le c_i \text{ for all } i \le n) \ge \prod_{i=1}^{n}\mathbb{P}(|\eta_i| \le c_i).$$

Theorem B (Anderson [And55, Theorem 1 and Corollary 2]). *Let η be a centered Gaussian vector in $\mathbb{R}^m$ and let $K \subset \mathbb{R}^m$ be closed, convex and symmetric ($K = -K$). Then $\mathbb{P}(\eta \in K + c) \le \mathbb{P}(\eta \in K)$ for every $c \in \mathbb{R}^m$.*

Both inequalities are applied to possibly singular covariances. Šidák proves his inequality for zero means, arbitrary variances and an arbitrary correlation matrix and obtains the singular case by approximation with nonsingular distributions [Š67, Theorem 1, Corollary 1 and p. 628]; coordinates with zero variance are trivial. Anderson proved his inequality for symmetric unimodal densities [And55, Theorem 1 and Corollary 2], which covers nondegenerate centered Gaussian vectors (see also [Bog98, Section 1.8]); the degenerate case follows by adding an independent noise δg with g standard Gaussian, applying the nondegenerate case to the enlarged set $K + [-\rho,\rho]^m$, and letting first $\delta \to 0$ and then $\rho \to 0$, using that K is closed.

Lemma 2.4 (Second moment of a conditioned Gaussian vector). *Let $K \subset \mathbb{R}^M$ be closed, convex and symmetric with $\gamma_M(K) > 0$, and let Z be standard Gaussian in $\mathbb{R}^M$. Then $\mathbb{E}(\|Z\|_2^2 \mid Z \in K) \le M$.*

Proof. Write $Z = R\Theta$ with $R := \|Z\|_2$ and $\Theta := Z/\|Z\|_2$; then R and Θ are independent and Θ is uniformly distributed on the unit sphere. Let $a(r) := \mathbb{P}(r\Theta \in K)$. Since K is convex, symmetric

and nonempty, $0 \in K$, so $r\omega \in K$ implies $r'\omega \in K$ for $0 \le r' \le r$ and every unit vector ω; hence a is nonincreasing. By independence, $\mathbb{E}(\|Z\|_2^2 \mathbf{1}_{\{Z\in K\}}) = \mathbb{E}(R^2 a(R))$, and Chebyshev's association inequality, applied to the nondecreasing function $r \mapsto r^2$ and the nonincreasing function a of the random variable R, gives $\mathbb{E}(R^2 a(R)) \le \mathbb{E}(R^2)\,\mathbb{E}(a(R)) = M\gamma_M(K)$. □

For a random vector $Z = (Z_1, \dots, Z_M)$ and $j \le M$ we write $Z_{<j} := (Z_1, \dots, Z_{j-1})$ and $Z_{\le j} := (Z_1, \dots, Z_j)$, and similarly $z_{<j}$ and $z_{\le j}$ for points $z \in \mathbb{R}^M$.

Lemma 2.5 (Rosenblatt transform [Ros52])**.** *Let $Z = (Z_1, \dots, Z_M)$ be a random vector with a density on $\mathbb{R}^M$. For $j \le M$ let $F_j(\cdot \mid z_{<j})$ be a regular version of the conditional distribution function of Z_j given $Z_{<j} = z_{<j}$ (the distribution function of Z_1 for $j = 1$), and put $U_j := 2F_j(Z_j \mid Z_{<j}) - 1$. Then $U_1, \dots, U_M$ are independent and uniformly distributed on $[-1, 1]$.*

Proof. For $\mathbb{P}_{Z_{<j}}$-almost every $z_{<j}$, where $\mathbb{P}_{Z_{<j}}$ is the law of $Z_{<j}$, the conditional law of Z_j given $Z_{<j} = z_{<j}$ has a density, so that $F_j(\cdot \mid z_{<j})$ is continuous and U_j is uniform on $[-1, 1]$ conditionally on $Z_{<j}$; hence U_j is uniform on $[-1, 1]$ and independent of $U_1, \dots, U_{j-1}$, which are functions of $Z_{<j}$. Induction on j gives the claim. □

2.5 Differential entropy

For a random vector Z with density φ on $\mathbb{R}^k$, the differential entropy is $\mathsf{h}(Z) := -\int \varphi \log \varphi$, whenever this integral exists; we also write $\mathsf{h}(\mu) := \mathsf{h}(Z)$ if Z has law μ. If φ is bounded with bounded support, then $\varphi \log \varphi$ is bounded and integrable, so $\mathsf{h}(Z)$ is finite; moreover all marginal densities of Z are bounded with bounded support. We use two standard facts.

For a random vector $Z = (Z_1, \dots, Z_M)$ with a bounded density with bounded support, let $\varphi_{\le j}$ and $\varphi_{<j}$ be the densities of $Z_{\le j}$ and $Z_{<j}$ (with $\varphi_{<1} := 1$), and let $\mathbb{P}_{Z_{<j}}$ be the law of $Z_{<j}$. For $\mathbb{P}_{Z_{<j}}$-almost every $z_{<j}$, namely whenever $\varphi_{<j}(z_{<j}) > 0$, the conditional law of Z_j given $Z_{<j} = z_{<j}$ has the bounded density $\varphi_{\le j}(z_{<j}, \cdot)/\varphi_{<j}(z_{<j})$ with bounded support; we write $\mathsf{h}(Z_j \mid Z_{<j} = z_{<j})$ for its entropy, which is finite, and $\mathsf{h}(Z_j \mid Z_{<j}) := \mathbb{E}(\mathsf{h}(Z_j \mid Z_{<j} = Z_{<j}))$ for its average, so that $\mathsf{h}(Z_1 \mid Z_{<1}) = \mathsf{h}(Z_1)$.

Lemma 2.6 (Chain rule)**.** *Let $Z = (Z_1, \dots, Z_M)$ have a bounded density with bounded support. Then each $\mathsf{h}(Z_j \mid Z_{<j})$ is finite and*

$$\mathsf{h}(Z) = \sum_{j=1}^{M} \mathsf{h}(Z_j \mid Z_{<j}).$$

Proof. For $z_{<j}$ with $\varphi_{<j}(z_{<j}) > 0$, the conditional density is $\varphi_{\le j}(z_{<j}, \cdot)/\varphi_{<j}(z_{<j})$, and

$$\mathsf{h}(Z_j \mid Z_{<j}) = -\int \varphi_{\le j} \log \varphi_{\le j} + \int \varphi_{\le j} \log \varphi_{<j} = \mathsf{h}(Z_{\le j}) - \mathsf{h}(Z_{<j}),$$

where Fubini's theorem applies because $\varphi_{\le j}|\log \varphi_{\le j}|$ and $\varphi_{<j}|\log \varphi_{<j}|$ are integrable. With $\mathsf{h}(Z_{<1}) := 0$, summing over j gives the claim; see also [CT06, Theorem 8.6.2]. □

Lemma 2.7 (Entropy and mean absolute deviation)**.** *Let Z be a real random variable with a density, finite entropy and $\mathbb{E}(|Z|) < \infty$, and let $\beta := \mathbb{E}(|Z - \mathbb{E}\,Z|)$, which is positive since Z has a density. Then $\mathsf{h}(Z) \le \log(2\mathrm{e}\beta)$.*

Proof. Let φ be the density of Z, $\bar{z} := \mathbb{E}\, Z$, and $\ell(t) := (2\beta)^{-1}\mathrm{e}^{-|t-\bar{z}|/\beta}$. Gibbs' inequality $\int \varphi \log(\varphi/\ell) \geq 0$ gives $\mathsf{h}(Z) \leq -\int \varphi \log \ell = \log(2\beta) + \beta^{-1}\,\mathbb{E}(|Z-\bar{z}|) = \log(2\mathrm{e}\beta)$. □

2.6 Sampling discretization

The points of our construction are provided by a sampling discretization theorem of Limonova and Temlyakov. We state the special case that we need: it is their Theorem 1.1 under their Condition E with parameter $t = 1$, which is the pointwise bound $\sum_k |u_k(x)|^2 \leq D$ below.

Theorem C (Limonova and Temlyakov [LT22, Theorem 1.1])**.** *There are absolute constants $C_1, C_2, C_3 > 0$ with the following property. Let μ be a probability measure on a nonempty set $\Omega \subset \mathbb{R}^d$, and let $u_1, \dots, u_D$ be real functions on Ω that are orthonormal in $L^2(\Omega, \mu)$ and satisfy $\sum_{k=1}^{D} |u_k(x)|^2 \leq D$ for every $x \in \Omega$. Then there are points $\xi^1, \dots, \xi^m \in \Omega$ with $m \leq C_1 D$ such that*

$$C_2 \,\|f\|^{\,2}_{L^2(\mu)} \leq \frac{1}{m}\sum_{j=1}^{m} |f(\xi^j)|^2 \leq C_3 \,\|f\|^{\,2}_{L^2(\mu)} \qquad \text{for all } f \in \operatorname{span}\{u_1, \dots, u_D\}.$$

The points ξ^j need not be distinct; distinct points are obtained in Lemma 5.3. The lower inequality implies $m \geq D$, since the evaluation map $f \mapsto (f(\xi^j))_{j \leq m}$ is injective on the D-dimensional space $\operatorname{span}\{u_1, \dots, u_D\}$; this is used in the proof of Lemma 5.3.

3 The test functions

We construct the test functions $W(u)$, $u \in [-1,1]^M$, of Step 3 in the overview, prove that they have L^1-norm at most one, and identify their linear part. The construction is parametrized by a depth $h \in 3\mathbb{N}$, and we write

$$v := 2^{-h}. \tag{3.1}$$

3.1 Shell boxes and localized Walsh functions

A *shape* is a vector $r \in (3\mathbb{N}_0)^d$ with $|r|_1 = h$; the set of shapes is $\mathcal{S}_h$, and its cardinality is

$$K_h := |\mathcal{S}_h| = \binom{h/3 + d - 1}{d-1}, \qquad c_d (h+1)^{d-1} \leq K_h \leq C_d (h+1)^{d-1}. \tag{3.2}$$

For $r \in \mathcal{S}_h$, the *shell boxes of shape* r are the dyadic boxes $R = R_1 \times \cdots \times R_d$, that is, products of dyadic intervals, with $R_j \in \mathcal{D}_{r_j}$; there are $2^h = 1/v$ of them, each of volume v. We write $\mathcal{R}$ for the set of all shell boxes, $\operatorname{shape}(R)$ for the shape of $R \in \mathcal{R}$, and

$$M := |\mathcal{R}| = K_h / v. \tag{3.3}$$

To every $R \in \mathcal{R}$ we attach the *localized Walsh function* q_R and its normalized primitive T_R,

$$q_R := q_{R_1} \otimes \cdots \otimes q_{R_d}, \qquad T_R := v^{-1}\, \psi_{R_1} \otimes \cdots \otimes \psi_{R_d}, \tag{3.4}$$

with q_I and ψ_I as in (2.5). Then $|q_R| = \mathbf{1}_R$, so $\|q_R\|_1 = v$ and $\|q_R\|_2^2 = v$. Moreover, q_R is a combination of tensor Haar functions of level $\sum_j (r_j + 2) = h + 2d$, and $J^* q_R = v T_R$.

Lemma 3.1 (Orthogonality of the primitives)**.** *For all $R, S \in \mathcal{R}$, $\langle T_R, T_S \rangle = \gamma_d v\, \mathbf{1}_{\{R=S\}}$, where $\gamma_d := 192^{-d}$.*

Proof. We have $\langle T_R, T_S\rangle = v^{-2}\prod_{j=1}^d \langle \psi_{R_j}, \psi_{S_j}\rangle$. If $R = S$, then (2.6) gives $v^{-2}\prod_j |R_j|^3/192 = v^{-2}\, v^3\, 192^{-d} = \gamma_d v$. If $R \neq S$ have the same shape, then $R_j \neq S_j$ for some j, these intervals have the same depth and are therefore disjoint, and $\langle \psi_{R_j}, \psi_{S_j}\rangle = 0$. If the shapes differ, there is a coordinate j with $r_j \neq s_j$, say $r_j < s_j$; then $s_j \geq r_j + 3$ because the shapes have coordinates in $3\mathbb{N}_0$. Either $S_j \cap R_j = \emptyset$, or S_j is contained in an interval of $\mathcal{D}_{r_j+3}$ inside R_j, on which ψ_{R_j} is affine by Lemma 2.2; in both cases $\langle \psi_{R_j}, \psi_{S_j}\rangle = 0$ by the last assertion of Lemma 2.2. □

3.2 Blocks and the positive product

The shapes are grouped according to their first coordinate. For $a \in 3\mathbb{N}_0$ with $a \leq h$, the *block* $\mathcal{C}_a := \{r \in \mathcal{S}_h : r_1 = a\}$ has cardinality

$$m_a := |\mathcal{C}_a| = \binom{(h-a)/3 + d - 2}{d-2} \leq (h+1)^{d-2}, \tag{3.5}$$

where the inequality follows from $\binom{n+k}{k} = \prod_{i=1}^k \frac{n+i}{i} \leq (n+1)^k$ and $(h-a)/3 + 1 \leq h+1$. There are $L_h := h/3 + 1$ blocks, indexed by $a \in \{0, 3, \dots, h\}$; we also refer to the index a itself as a block. We fix the parameters

$$\varepsilon := \tfrac{1}{4}, \qquad \theta := t_d\,(h+1)^{-(d-2)/2}, \qquad \lambda := \varepsilon\theta, \tag{3.6}$$

where $t_d \in (0, \frac{1}{2}]$ is a constant depending only on d; it is chosen in Section 4, see (4.14). By (3.5),

$$\theta^2 m_a \leq t_d^2 \leq \tfrac{1}{4} \qquad \text{for every block } a. \tag{3.7}$$

Thus the sign sums $\theta\sum_{r\in\mathcal{C}_a}\xi_r$ defining the functions Φ_a below have variance at most $\frac{1}{4}$; Lemma 2.3 controls the resulting nonlinear Fourier–Walsh terms.

A *coefficient vector* is a vector $u = (u_R)_{R\in\mathcal{R}} \in [-1,1]^{\mathcal{R}}$; we fix an enumeration of $\mathcal{R}$ and identify $[-1,1]^{\mathcal{R}}$ with $[-1,1]^M$ and $\mathbb{R}^{\mathcal{R}}$ with $\mathbb{R}^M$. For a shape r, the *layer function*

$$f_r(u;x) := \sum_{R\in\mathcal{R}:\ \mathrm{shape}(R)=r} u_R\, q_R(x), \qquad x \in [0,1]^d, \tag{3.8}$$

satisfies $|f_r(u;x)| \leq 1$, because the shell boxes of shape r are disjoint and $|q_R| = \mathbf{1}_R$. For a block a, let $\Phi_a\colon \{-1,1\}^{\mathcal{C}_a} \to [-1,1]$, $\Phi_a(\xi) := \tanh\bigl(\theta\sum_{r\in\mathcal{C}_a}\xi_r\bigr)$, and let $\widetilde{\Phi}_a$ be its multilinear extension (2.7). The *block function* is

$$B_a(u;x) := \widetilde{\Phi}_a\bigl((f_r(u;x))_{r\in\mathcal{C}_a}\bigr) = \sum_{T\subseteq\mathcal{C}_a} \widehat{\Phi}_a(T)\prod_{r\in T} f_r(u;x), \tag{3.9}$$

and $|B_a(u;x)| \leq 1$ by the remark following (2.7). Finally, we define the *positive products* and the *test function*

$$Q_\pm(u;x) := \prod_a \bigl(1 \pm \varepsilon B_a(u;x)\bigr), \qquad W(u;x) := \tfrac{1}{2}\bigl(Q_+(u;x) - Q_-(u;x)\bigr), \tag{3.10}$$

where the products run over all L_h blocks. Since $\varepsilon|B_a| \leq 1/4$, we have $Q_\pm \geq (3/4)^{L_h} > 0$. Expanding the products,

$$W(u;x) = \sum_{\substack{S\subseteq\{0,3,\dots,h\}\\ |S| \text{ odd}}} \varepsilon^{|S|}\prod_{a\in S} B_a(u;x). \tag{3.11}$$

3.3 Normalization

The following lemma is the reason for the product structure of W and for grouping the shapes by their first coordinate: integrating along the first coordinate removes one block at a time.

Lemma 3.2 (Normalization)**.** *For every $u \in [-1,1]^{\mathcal{R}}$ and every $x' = (x_2, \dots, x_d) \in [0,1)^{d-1}$,*

$$\int_0^1 Q_\pm(u; x_1, x')\, dx_1 = 1.$$

Consequently, $\|Q_\pm(u)\|_1 = 1$ and $\|W(u)\|_1 \le 1$ for every $u \in [-1,1]^{\mathcal{R}}$.

Proof. Fix u and x', and consider all functions as functions of $x_1 \in [0,1)$. Let a be a block and $r \in \mathcal{C}_a$. For $x_1 \in I \in \mathcal{D}_a$, the unique shell box R of shape r containing (x_1, x') is $R = I \times R_2 \times \cdots \times R_d$, where $R_j \in \mathcal{D}_{r_j}$ is the interval containing x_j; these R_j do not depend on x_1. Hence

$$f_r(u; x_1, x') = q_I(x_1)\, c_r(I), \qquad c_r(I) := u_R \prod_{j=2}^{d} q_{R_j}(x_j) \in [-1,1], \qquad x_1 \in I,$$

and $c_r(I)$ depends on x_1 only through I. By Lemma 2.3(i), only sets T of odd cardinality occur in (3.9), and $q_I^{|T|} = q_I$ on I for odd $|T|$. Therefore, for $x_1 \in I$,

$$B_a(u; x_1, x') = q_I(x_1)\, \zeta_a(I), \qquad \zeta_a(I) := \sum_{T \subseteq \mathcal{C}_a} \widehat{\Phi}_a(T) \prod_{r \in T} c_r(I) = \widetilde{\Phi}_a\big((c_r(I))_{r \in \mathcal{C}_a}\big) \in [-1,1].$$

Since $q_I = \mathbf{1}_I \varrho_{a+1} \varrho_{a+2} \varrho_{a+3}$, we conclude that

$$B_a(u; \cdot, x') = \varrho_{a+3}\, g_a, \qquad g_a := \varrho_{a+1} \varrho_{a+2} \sum_{I \in \mathcal{D}_a} \zeta_a(I) \mathbf{1}_I, \tag{3.12}$$

where g_a is constant on every interval of $\mathcal{D}_{a+2}$. In particular, $B_a(u; \cdot, x')$ is constant on every interval of $\mathcal{D}_{a+3}$, and $\int_J B_a(u; x_1, x')\, dx_1 = 0$ for every $J \in \mathcal{D}_{a+2}$, because ϱ_{a+3} has integral zero over J.

We remove the factors of $Q_\pm$ one at a time. Let a be the largest block index among the remaining factors. Every other remaining block a' satisfies $a' + 3 \le a$, so its factor $1 \pm \varepsilon B_{a'}$ is constant on the intervals of $\mathcal{D}_a$ and hence on those of $\mathcal{D}_{a+2}$, while $\int_J (1 \pm \varepsilon B_a)\, dx_1 = |J|$ for every $J \in \mathcal{D}_{a+2}$. Integrating over each $J \in \mathcal{D}_{a+2}$ and summing, we see that the integral over $[0,1)$ of the product of the remaining factors does not change when the factor $1 \pm \varepsilon B_a$ is removed. Continuing with the remaining factors until the empty product 1 is reached gives $\int_0^1 Q_\pm(u; x_1, x')\, dx_1 = 1$. As $Q_\pm > 0$, Fubini's theorem yields $\|Q_\pm(u)\|_1 = 1$, and $\|W(u)\|_1 \le \frac{1}{2}(\|Q_+(u)\|_1 + \|Q_-(u)\|_1) = 1$. □

3.4 The linear part

Each $B_a(u; x)$ is a polynomial in the coefficients u, and so is $W(u; x)$; we isolate the part of $W(u; x)$ of degree one in u. By Lemma 2.3(ii) with $m = m_a$, the first-order Fourier–Walsh coefficient $b_a := \widehat{\Phi}_a(\{r\})$, $r \in \mathcal{C}_a$, does not depend on r. Put

$$\lambda_R := \varepsilon\, b_a \qquad \text{for } R \in \mathcal{R} \text{ with } \operatorname{shape}(R) \in \mathcal{C}_a, \tag{3.13}$$

and define

$$W_{\mathrm{lin}}(u) := \sum_{R \in \mathcal{R}} \lambda_R u_R q_R, \qquad F_{\mathrm{lin}}(u) := J^* W_{\mathrm{lin}}(u) = v \sum_{R \in \mathcal{R}} \lambda_R u_R T_R, \tag{3.14}$$

where $F_{\mathrm{lin}} = J^* W_{\mathrm{lin}}$ follows from $J^* q_R = v T_R$.

Lemma 3.3 (Linear part)**.** *For every block a, $\theta/2 \le b_a \le \theta$; hence $\frac{1}{2}\lambda \le \lambda_R \le \lambda$ for every $R \in \mathcal{R}$. The difference $W(u) - W_{\mathrm{lin}}(u)$ is a polynomial in u all of whose monomials have degree at least three. Moreover, if U is uniformly distributed on $[-1,1]^{\mathcal{R}}$, then*

$$\begin{aligned} \mathbb{E}_U\big(\|F_{\mathrm{lin}}(U)\|_2^2\big) &= \frac{\gamma_d v^3}{3} \sum_{R\in\mathcal{R}} \lambda_R^2, \\ c_d\lambda^2(h+1)^{d-1}v^2 \le\ \mathbb{E}_U\big(\|F_{\mathrm{lin}}(U)\|_2^2\big) &\le C_d\lambda^2(h+1)^{d-1}v^2. \end{aligned} \tag{3.15}$$

Proof. Lemma 2.3(ii) with $m = m_a$ gives $b_a \le \theta$ and $b_a \ge \theta\big(1-\theta^2(m_a-\frac{2}{3})\big) \ge \theta(1-\theta^2 m_a) \ge \theta/2$ by (3.7). The layer functions $f_r(u)$ are linear in u, and distinct layer functions involve disjoint sets of coefficients. By (3.9) and Lemma 2.3(i), the part of $B_a(u)$ of degree one in u is $b_a\sum_{r\in\mathcal{C}_a} f_r(u)$, and all other monomials of $B_a(u)$ have odd degree at least three. In (3.11), a monomial of degree one can only arise from a singleton $S=\{a\}$ and the degree-one part of B_a; this gives $\sum_a \varepsilon b_a \sum_{r\in\mathcal{C}_a} f_r(u) = W_{\mathrm{lin}}(u)$. All remaining monomials have degree at least three. Finally, since $\mathbb{E}(U_R U_S) = \frac{1}{3}\mathbf{1}_{\{R=S\}}$, Lemma 3.1 gives $\mathbb{E}_U(\|F_{\mathrm{lin}}(U)\|_2^2) = v^2\sum_R \lambda_R^2 \cdot \frac{1}{3}\gamma_d v$, and (3.15) follows from $\lambda/2 \le \lambda_R \le \lambda$, $M = K_h/v$ and (3.2). □

Remark 3.4 (Depth separation and smoothing)**.** The depths of the shell boxes are multiples of three, so that two different depths in the same coordinate differ by at least three. Since q_I is constant on every dyadic interval of depth at least $\mathrm{depth}(I)+3$, every product of functions q_I in one coordinate is therefore either zero or, up to sign, equal to $q_{I'}$ or $\mathbf{1}_{I'}$ for the deepest interval I' involved, according to the parity of the number of factors of that depth (Lemma 4.2); likewise, the proof of Lemma 3.2 can peel off one block at a time. The smoothing tanh in the block functions keeps $|B_a| \le 1$, and hence $1 \pm \varepsilon B_a \ge \frac{3}{4}$, for all coefficient vectors and all points, while by Lemma 2.3 the function Φ_a is linear up to cubic and higher Fourier–Walsh coefficients whose squares sum to at most $\theta^6 m_a^3$.

4 The nonlinear remainder

The test function $W(u)$ is far from linear in u as a function on the cube. In this section we show that it is linear up to a small relative error once its component in the low Haar cross is removed and the primitive is taken. Let

$$\mathcal{H}_{\mathrm{lo}} := \mathrm{span}\big\{\text{tensor Haar functions of level at most } h-1\big\} \subset L^2([0,1]^d), \tag{4.1}$$

a finite-dimensional space, which we call the *low Haar cross*, and let P_{lo} be the orthogonal projection of $L^2([0,1]^d)$ onto $\mathcal{H}_{\mathrm{lo}}$ and $P_{\mathrm{hi}} := \mathrm{Id} - P_{\mathrm{lo}}$. Since q_R is a combination of tensor Haar functions of level $h+2d$, we have

$$P_{\mathrm{lo}} W_{\mathrm{lin}}(u) = 0 \qquad \text{and} \qquad P_{\mathrm{hi}} W_{\mathrm{lin}}(u) = W_{\mathrm{lin}}(u) \qquad \text{for every } u. \tag{4.2}$$

The component in the low Haar cross has to be removed because of coincidences of depths. If, in a coordinate $k \ge 2$, the largest side depth among the shell boxes in a product $\prod_R q_R$ is attained by an even number of them, then the factors in that coordinate multiply to $\pm\mathbf{1}_I$, where I is the common deepest side, because $q_I^2 = \mathbf{1}_I$ and the factors of smaller depth are constant on I (Lemma 4.2). The primitive of q_I vanishes outside I, but the primitive of $\mathbf{1}_I$ equals $|I|$ to the left of I, so the primitive of such a product does not vanish outside its support; the part far from I comes from the constant and the coarse Haar components of $\mathbf{1}_I$. The projection P_{hi} removes all tensor Haar components of

level below h. It therefore retains only components whose depths fall short of the depths of the deepest sides by at most $e + 2d$ in total, where e is the excess of the join of the shapes involved (Section 4.1), and Lemma 4.3 turns this into exponential decay in e.

Theorem 4.1 (Remainder estimate)**.** *There is a constant $t_d^* \in (0, \frac{1}{2}]$ depending only on d such that the following holds for every $t_d \in (0, t_d^*]$ in (3.6) and every $h \in 3\mathbb{N}$, with constants C_d, C_d' that depend only on d and not on t_d. If U is uniformly distributed on $[-1,1]^{\mathcal{R}}$, then*

$$\mathbb{E}_U\big(\|J^* P_{\mathrm{hi}} W(U) - F_{\mathrm{lin}}(U)\|_2^2\big) \le C_d\, \lambda^2 (h+1)^{d-1} v^2\, (\theta^4 + \lambda^4) \le C_d'\, \theta^4\ \mathbb{E}_U\big(\|F_{\mathrm{lin}}(U)\|_2^2\big). \tag{4.3}$$

The second inequality follows from the first by (3.15), whose constants do not depend on t_d, and $\lambda \le \theta$. Since $\theta^4 = t_d^4 (h+1)^{-2(d-2)}$, the relative root-mean-square error is $O_d((h+1)^{-(d-2)})$, which tends to zero as $h \to \infty$ because $d \ge 3$.

From now on we fix $t_d := t_d^*$, with t_d^* as chosen in (4.14); the constants c_d, C_d appearing later may depend on this choice, whereas the constants in (4.3) do not.

Strategy. We expand $W(u)$ into monomials in the coefficients and use their orthogonality to reduce the mean-square error to a sum of squared norms. After removal of the component in the low Haar cross, the cell-location estimate gives exponential decay in the join excess, up to polynomial factors. We then separate monomials lying in one block from those meeting at least three blocks; their total contributions are bounded by factors $O_d(\theta^4)$ and $O_d(\lambda^4)$, respectively, relative to the mean-square size of the linear part.

4.1 Cell monomials and joins

Substituting (3.9) and (3.8) into (3.11) and expanding all products, we write $W(u)$ as a sum of monomials in the coefficients. Each term is indexed by a set $\Gamma \subseteq \mathcal{R}$ of shell boxes with pairwise distinct shapes, one from each layer function that occurs; we write $\mathcal{T}(\Gamma) := \{\mathrm{shape}(R) : R \in \Gamma\}$ for its set of shapes and $S(\Gamma) := \{r_1 : r \in \mathcal{T}(\Gamma)\}$ for the set of blocks it meets. We call Γ *admissible* if $|S(\Gamma)|$ is odd and $|\mathcal{T}(\Gamma) \cap \mathcal{C}_a|$ is odd for every $a \in S(\Gamma)$, and we denote the set of admissible Γ by $\mathcal{M}$. For $\Gamma \in \mathcal{M}$ we put

$$c_\Gamma := \varepsilon^{|S(\Gamma)|} \prod_{a \in S(\Gamma)} \widehat{\Phi}_a\big(\mathcal{T}(\Gamma) \cap \mathcal{C}_a\big), \qquad m_\Gamma(u) := \prod_{R \in \Gamma} u_R, \qquad X_\Gamma := \prod_{R \in \Gamma} q_R. \tag{4.4}$$

Then, by Lemma 2.3(i), which shows that only odd sets contribute in (3.9),

$$W(u) = \sum_{\Gamma \in \mathcal{M}} c_\Gamma\, m_\Gamma(u)\, X_\Gamma, \qquad W_{\mathrm{lin}}(u) = \sum_{\Gamma \in \mathcal{M},\ |\Gamma| = 1} c_\Gamma\, m_\Gamma(u)\, X_\Gamma, \tag{4.5}$$

where the second identity restates Lemma 3.3; note that $|\Gamma| = 1$ means $|S(\Gamma)| = 1$ and $|\mathcal{T}(\Gamma)| = 1$, and that every other admissible Γ has $|\Gamma| \ge 3$. We call the terms $c_\Gamma m_\Gamma(u) X_\Gamma$ with $\Gamma \in \mathcal{M}$ the *cell monomials* of W; for $X_\Gamma \ne 0$, the dyadic box $\bigcap_{R \in \Gamma} R$, on which X_Γ is supported, is called the *cell* of Γ. Each m_Γ is squarefree, and distinct admissible sets give distinct monomials. If U is uniformly distributed on $[-1,1]^{\mathcal{R}}$, then $\mathbb{E}(m_\Gamma(U) m_{\Gamma'}(U)) = 0$ for $\Gamma \ne \Gamma'$, because some coefficient occurs exactly once in the product, and $\mathbb{E}(m_\Gamma(U)^2) = 3^{-|\Gamma|} \le 1$. Consequently, for every bounded linear operator $\mathcal{O}$ on $L^2([0,1]^d)$,

$$\mathbb{E}_U\big(\big\|\mathcal{O}\big(W(U) - W_{\mathrm{lin}}(U)\big)\big\|_2^2\big) = \sum_{\Gamma \in \mathcal{M},\ |\Gamma| \ge 3} c_\Gamma^2\, 3^{-|\Gamma|}\, \|\mathcal{O} X_\Gamma\|_2^2 \le \sum_{\Gamma \in \mathcal{M},\ |\Gamma| \ge 3} c_\Gamma^2\ \|\mathcal{O} X_\Gamma\|_2^2. \tag{4.6}$$

We apply this with $\mathcal{O} = J^* P_{\mathrm{hi}}$, which is bounded since J^* is bounded on $L^2([0,1]^d)$; by (4.2), the left-hand side of (4.6) is then the left-hand side of (4.3). To bound the right-hand side of (4.6), we describe the functions X_Γ.

For a set of shapes $\mathcal{T} \subseteq \mathcal{S}_h$, the *join* $j = j(\mathcal{T}) \in (3\mathbb{N}_0)^d$ is the coordinatewise maximum $j_k := \max_{r \in \mathcal{T}} r_k$, and the *excess* is $e(\mathcal{T}) := |j|_1 - h$. Since $|r|_1 = h$ for every shape and $r_k \le h$, we have $0 \le e(\mathcal{T}) \le (d-1)h$.

Lemma 4.2 (Structure of cell monomials)**.** *Let $\mathcal{T} = \mathcal{T}(\Gamma_0)$ for some $\Gamma_0 \in \mathcal{M}$, with join j and excess e.*

(a) *The admissible sets Γ with $\mathcal{T}(\Gamma) = \mathcal{T}$ and $X_\Gamma \neq 0$ are in bijection with the dyadic boxes $I = I_1 \times \cdots \times I_d$ with $I_k \in \mathcal{D}_{j_k}$, via $\Gamma \mapsto I := \bigcap_{R \in \Gamma} R$. In particular there are exactly 2^{h+e} of them.*

(b) *If Γ corresponds to I in* (a)*, then*

$$X_\Gamma = \pm\, q_{I_1} \otimes g_2 \otimes \cdots \otimes g_d, \qquad g_k = \begin{cases} q_{I_k} & \text{if } \#\{r \in \mathcal{T} : r_k = j_k\} \text{ is odd,} \\ \mathbf{1}_{I_k} & \text{otherwise.} \end{cases} \tag{4.7}$$

Proof. (a) If $X_\Gamma \neq 0$, the boxes in Γ have a common point, so in every coordinate k their sides are pairwise nested, and their intersection is a dyadic box I whose k-th side is the deepest of these sides, of depth $\max_{r \in \mathcal{T}} r_k = j_k$. Conversely, given a dyadic box I with sides of depths j_k, every shape $r \in \mathcal{T}$ satisfies $r \le j$, and the unique shell box R_r of shape r containing I is the product of the ancestors of the sides of I of depths r_k. The set $\Gamma := \{R_r : r \in \mathcal{T}\}$ is admissible, has $\mathcal{T}(\Gamma) = \mathcal{T}$ and $\bigcap_{R \in \Gamma} R = I$, and $X_\Gamma \neq 0$ because $|X_\Gamma| = \prod_{R \in \Gamma} \mathbf{1}_R = \mathbf{1}_I$; the two maps are inverse to each other. The number of dyadic boxes with sides of depths $j_1, \dots, j_d$ is $2^{|j|_1} = 2^{h+e}$.

(b) Write R_r for the shell box of shape r in Γ and $R_{r,k}$ for its k-th side, so that $X_\Gamma = \bigotimes_{k=1}^d \prod_{r \in \mathcal{T}} q_{R_{r,k}}$. Fix a coordinate k. The product $\prod_{r \in \mathcal{T}} q_{R_{r,k}}$ vanishes outside I_k, since it contains the factor q_{I_k} (some r has $r_k = j_k$). Let $x_k \in I_k$. For $r \in \mathcal{T}$ with $r_k < j_k$ we have $r_k \le j_k - 3$, so I_k is contained in an interval of $\mathcal{D}_{r_k+3}$ inside $R_{r,k}$, on which $q_{R_{r,k}}$ is constant and equal to ± 1. For $r \in \mathcal{T}$ with $r_k = j_k$ we have $R_{r,k} = I_k$ and the factor is q_{I_k}. Since $q_{I_k}^2 = \mathbf{1}_{I_k}$, the product $\prod_{r \in \mathcal{T}} q_{R_{r,k}}(x_k)$ equals $\pm q_{I_k}(x_k)$ or $\pm \mathbf{1}_{I_k}(x_k)$ according to the parity of $\#\{r : r_k = j_k\}$. In the first coordinate, $j_1 = \max S(\Gamma)$ is the block of largest index met by Γ, and the shapes with $r_1 = j_1$ are exactly the elements of $\mathcal{T} \cap \mathcal{C}_{j_1}$, whose number is odd by admissibility. □

4.2 The cell-location estimate

The following lemma bounds the total contribution of all monomials with a fixed set of shapes.

Lemma 4.3 (Cell-location estimate)**.** *Let $\mathcal{T}$ be as in Lemma 4.2, with excess e. Then*

$$\sum_{\Gamma \in \mathcal{M}:\ \mathcal{T}(\Gamma) = \mathcal{T}} \| J^* P_{\mathrm{hi}} X_\Gamma \|_2^2 \le C_d\, v^2\, 2^{-e}\, (e+1)^{2(d-1)}.$$

Proof. Sets Γ with $X_\Gamma = 0$ contribute nothing. Let Γ with $X_\Gamma \neq 0$ correspond to the box I in Lemma 4.2(a), and write X_Γ as in (4.7). Call a coordinate $k \ge 2$ *modulated* if $g_k = q_{I_k}$ and *plain* otherwise. We expand every factor in the Haar basis. The factor q_{I_1} and every modulated factor q_{I_k}

are combinations of four Haar functions of level j_1+2 and j_k+2, respectively. A plain factor $\mathbf{1}_{I_k}$ has the expansion

$$\mathbf{1}_{I_k} = 2^{-j_k}\mathbf{1} + \sum_{b=0}^{j_k-1} \sigma_b\, 2^{b-j_k}\, h_{J_b}, \tag{4.8}$$

where J_b is the ancestor of I_k of depth b and $\sigma_b \in \{-1,1\}$ records on which half of J_b the interval I_k lies. For a *component vector* $\beta = (\beta_2, \dots, \beta_d)$ with $\beta_k = j_k$ for modulated k and $\beta_k \in \{-1, 0, \dots, j_k - 1\}$ for plain k, let $g_k^{(\beta_k)}$ be the corresponding term: $g_k^{(j_k)} := q_{I_k}$ for modulated k, and $g_k^{(-1)} := 2^{-j_k}\mathbf{1}$, $g_k^{(b)} := \sigma_b 2^{b-j_k} h_{J_b}$ for plain k. Then

$$X_\Gamma = \pm \sum_{\beta} X_\Gamma^{(\beta)}, \qquad X_\Gamma^{(\beta)} := q_{I_1} \otimes g_2^{(\beta_2)} \otimes \cdots \otimes g_d^{(\beta_d)},$$

and $X_\Gamma^{(\beta)}$ is a combination of tensor Haar functions of the common level

$$\operatorname{lev}(\beta) = (j_1+2) + \sum_{k\ge 2} \beta_k + 2\,\#\{k \ge 2 : k \text{ modulated}\}.$$

Hence $P_{\mathrm{hi}} X_\Gamma = \pm \sum_{\beta:\ \operatorname{lev}(\beta)\ge h} X_\Gamma^{(\beta)}$, and every retained component vector satisfies

$$j_1 + \sum_{k\ge 2} \beta_k \;\ge\; h - 2d. \tag{4.9}$$

The deficits $\delta_k := j_k - \beta_k \ge 0$ of a retained component vector satisfy $\sum_{k\ge2}\delta_k = (h+e-j_1) - \sum_{k\ge2}\beta_k \le e + 2d$ by (4.9). The number $N_{\mathcal{T}}$ of retained component vectors is therefore at most the number of vectors in $\mathbb{N}_0^{d-1}$ with coordinate sum at most $e+2d$, so that

$$N_{\mathcal{T}} \le \binom{e+3d-1}{d-1} \le C_d (e+1)^{d-1}. \tag{4.10}$$

Next we bound the primitives. By (2.6) and (2.4), $\|J^* q_{I_1}\|_2^2 = 2^{-3j_1}/192$, and for $k \ge 2$

$$\left\|J^* g_k^{(j_k)}\right\|_2^2 = \frac{2^{-3j_k}}{192}, \qquad \left\|J^* g_k^{(-1)}\right\|_2^2 = \frac{2^{-2j_k}}{3}, \qquad \left\|J^* g_k^{(b)}\right\|_2^2 = 2^{2b-2j_k}\cdot\frac{2^{-3b}}{12}, \qquad 0 \le b < j_k,$$

so that in all three cases $\|J^* g_k^{(\beta_k)}\|_2^2 \le 2^{-2j_k-\beta_k}$. Since J^* acts coordinatewise on tensor products,

$$\left\|J^* X_\Gamma^{(\beta)}\right\|_2^2 \le 2^{-3j_1 - \sum_{k\ge2}(2j_k+\beta_k)} = 2^{-2(h+e)-j_1-\sum_{k\ge2}\beta_k} \le 2^{2d}\, 2^{-3h-2e} = 2^{2d}\, v^3\, 2^{-2e},$$

where we used $\sum_{k\ge2} 2j_k = 2(h+e) - 2j_1$ and (4.9). This bound does not depend on the box I. The components $X_\Gamma^{(\beta)}$ for different β need not have orthogonal primitives, so we use the Cauchy–Schwarz inequality:

$$\|J^* P_{\mathrm{hi}} X_\Gamma\|_2^2 \le N_{\mathcal{T}} \sum_{\beta:\ \operatorname{lev}(\beta)\ge h} \left\|J^* X_\Gamma^{(\beta)}\right\|_2^2 \le N_{\mathcal{T}}^2\, 2^{2d}\, v^3\, 2^{-2e}.$$

Summing over the $2^{h+e} = v^{-1}2^e$ admissible sets Γ with $\mathcal{T}(\Gamma) = \mathcal{T}$ and $X_\Gamma \ne 0$ (the others contribute nothing) and using (4.10) gives the claim. □

4.3 Proof of Theorem 4.1

We first record two counting bounds.

Lemma 4.4 (Counting shapes and joins)**.** *Let $j \in (3\mathbb{N}_0)^d$ with $|j|_1 = h+e$, where $0 \le e \le (d-1)h$.*

(a) *The number of shapes $r \in \mathcal{S}_h$ with $r \le j$ is at most $C_d(e+1)^{d-1}$, and the number of those with $r_1 = j_1$ is at most $C_d(e+1)^{d-2}$.*

(b) *The number of vectors $j' \in (3\mathbb{N}_0)^d$ with $|j'|_1 = h+e$ is at most $C_d(h+1)^{d-1}$.*

Proof. (a) A shape $r \le j$ is determined by the vector $j - r \in (3\mathbb{N}_0)^d$, whose coordinate sum is e. The number of such vectors is at most $\binom{e/3+d-1}{d-1} \le C_d(e+1)^{d-1}$, and at most $\binom{e/3+d-2}{d-2} \le C_d(e+1)^{d-2}$ if the first coordinate is prescribed to be zero.

(b) The number of such j' is $\binom{(h+e)/3+d-1}{d-1} \le C_d(h+e+1)^{d-1} \le C_d(h+1)^{d-1}$, since $e \le (d-1)h$. □

Proof of Theorem 4.1. By (4.6) with $\mathcal{O} = J^*P_{\mathrm{hi}}$, the identity (4.2), and the fact that c_Γ depends only on $\mathcal{T}(\Gamma)$,

$$\begin{aligned}\mathbb{E}_U\big(\|J^*P_{\mathrm{hi}}W(U) - F_{\mathrm{lin}}(U)\|_2^{\,2}\big) &\le \sum_{\mathcal{T}} c_{\mathcal{T}}^2 \sum_{\Gamma\in\mathcal{M}:\ \mathcal{T}(\Gamma)=\mathcal{T}} \|J^*P_{\mathrm{hi}}X_\Gamma\|_2^{\,2} \\ &\le \sum_{\mathcal{T}} c_{\mathcal{T}}^2\, C_d v^2\, 2^{-e(\mathcal{T})}\,(e(\mathcal{T})+1)^{2(d-1)},\end{aligned} \tag{4.11}$$

where $\mathcal{T}$ ranges over the sets of shapes of admissible sets with at least three elements, $c_{\mathcal{T}}$ denotes the common value of c_Γ for $\mathcal{T}(\Gamma) = \mathcal{T}$, and we used Lemma 4.3. We group the sets $\mathcal{T}$ according to their join j and distinguish two classes. Since $|S(\Gamma)|$ is odd for admissible Γ, either $\mathcal{T}$ meets a single block, and then $|\mathcal{T}| \ge 3$ because $|\Gamma| \ge 3$ (class I), or it meets at least three blocks (class II).

Class I: one block. Here $\mathcal{T} \subseteq \mathcal{C}_a$ for a single block a, with $|\mathcal{T}| \ge 3$ odd, and $c_{\mathcal{T}} = \varepsilon\widehat{\Phi}_a(\mathcal{T})$. The join j of $\mathcal{T}$ satisfies $j_1 = a$, and $\mathcal{T}$ is a subset of $\mathcal{I}_j := \{r \in \mathcal{C}_{j_1} : r \le j\}$, whose cardinality is at most $C_d(e+1)^{d-2}$ by Lemma 4.4(a). By the second inequality in (2.9), applied to $F = \Phi_{j_1}$ and $\mathcal{E} = \mathcal{I}_j$,

$$\sum_{\mathcal{T}\subseteq\mathcal{I}_j,\ |\mathcal{T}|\ge 3} c_{\mathcal{T}}^2 \le \varepsilon^2\theta^6|\mathcal{I}_j|^3 \le C_d\,\varepsilon^2\theta^6(e+1)^{3(d-2)}.$$

For each excess e there are at most $C_d(h+1)^{d-1}$ joins by Lemma 4.4(b), and every excess satisfies $0 \le e \le (d-1)h$. Hence the contribution of class I to the right-hand side of (4.11) is at most

$$\begin{aligned}&\sum_{e=0}^{(d-1)h} C_d(h+1)^{d-1}\cdot\varepsilon^2\theta^6(e+1)^{3(d-2)}\cdot v^2 2^{-e}(e+1)^{2(d-1)} \\ &\quad\le C_d\,\varepsilon^2\theta^6(h+1)^{d-1}v^2\sum_{e\ge 0}2^{-e}(e+1)^{5d-8} \le C_d\,\lambda^2\theta^4(h+1)^{d-1}v^2.\end{aligned} \tag{4.12}$$

Class II: at least three blocks. Here $|S| \ge 3$, where S is the set of blocks met by $\mathcal{T}$, and $c_{\mathcal{T}}^2 = \prod_{a\in S}\varepsilon^2\widehat{\Phi}_a(\mathcal{T}\cap\mathcal{C}_a)^2$. Fix a join j with excess e and put $\mathcal{I}'_j := \{r \in \mathcal{S}_h : r \le j\}$, so that $|\mathcal{I}'_j| \le C_d(e+1)^{d-1}$ by Lemma 4.4(a). For a block a let

$$\alpha_a(j) := \varepsilon^2 \sum_{\emptyset\ne T\subseteq\mathcal{C}_a\cap\mathcal{I}'_j} \widehat{\Phi}_a(T)^2 \le \varepsilon^2\theta^2\,|\mathcal{C}_a\cap\mathcal{I}'_j|, \qquad \alpha(j) := \sum_a \alpha_a(j) \le \varepsilon^2\theta^2|\mathcal{I}'_j| \le C_d\,\lambda^2(e+1)^{d-1},$$

where we used the first inequality in (2.9). Every class II set $\mathcal{T}$ with join j has the form $\bigcup_{a\in S} T_a$ with nonempty $T_a \subseteq \mathcal{C}_a \cap \mathcal{I}'_j$. Summing $c_{\mathcal{T}}^2$ over all such choices with a fixed S gives at most $\prod_{a\in S}\alpha_a(j)$, and summing over all S with $|S|\ge 3$ gives at most

$$\sum_{k\ge 3}\sum_{|S|=k}\prod_{a\in S}\alpha_a(j) \le \sum_{k\ge 3}\frac{\alpha(j)^k}{k!} \le \frac{\alpha(j)^3}{6}\mathrm{e}^{\alpha(j)},$$

because the k-th elementary symmetric polynomial of nonnegative numbers is at most the k-th power of their sum divided by $k!$. Using Lemma 4.4(b) once more and recalling that the excess of every set of shapes satisfies $0 \le e \le (d-1)h$, the contribution of class II to (4.11) is at most

$$C_d(h+1)^{d-1}v^2 \sum_{e=0}^{(d-1)h} 2^{-e}(e+1)^{2(d-1)}\cdot\lambda^6(e+1)^{3(d-1)}\exp\bigl(C_d\lambda^2(e+1)^{d-1}\bigr). \tag{4.13}$$

For every e in this range, $e+1 \le (d-1)h+1 \le d(h+1)$, so the definition (3.6) of θ and λ gives

$$\lambda^2(e+1)^{d-1} = \varepsilon^2 t_d^2\,(h+1)^{-(d-2)}(e+1)^{d-2}\,(e+1) \le \varepsilon^2 t_d^2\, d^{\,d-2}\,(e+1).$$

We now choose

$$t_d^* \in (0,\tfrac12] \qquad \text{with} \qquad C_d\,\varepsilon^2 (t_d^*)^2 d^{\,d-2} \le \tfrac14 \log 2, \tag{4.14}$$

where C_d is the constant in the exponent in (4.13); it comes from Lemma 4.4(a) and does not depend on t_d. Then, for every $t_d \in (0, t_d^*]$, the exponential factor is at most $2^{(e+1)/4}$ for all $0 \le e \le (d-1)h$, so the sum in (4.13), without the factor λ^6, is at most $\sum_{e\ge 0} 2^{-e}(e+1)^{5(d-1)}2^{(e+1)/4}$, a convergent series whose value depends only on d; hence the contribution of class II is at most $C_d\lambda^6(h+1)^{d-1}v^2$. Together with (4.12) this proves the first inequality in (4.3). □

5 Sampling and the exact linear response

We choose the point set and the matrix B that transforms a standard Gaussian vector $z \in \mathbb{R}^M$ into the weight vector $\eta = Bz \in \mathbb{R}^n$. Three properties are needed: the linear part of the pairing of the weighted counting function G_z with $W(u)$ is exactly diagonal in u; the pairing does not see the low Haar cross, so that Theorem 4.1 controls the nonlinear remainder; and $BB^* \preceq I_n$, which gives the variance bound required in Lemma 7.1. The first two properties rest on a weighted point set for which discrete and continuous L^2-norms are equivalent on a space containing all T_R and the primitives of the low Haar cross (Section 5.1), and which also controls the mean-square remainder at the points (Section 5.2); Section 5.3 then defines B by a projection and an inverse Gram matrix.

5.1 A space of piecewise multiaffine functions

For $j \in \mathbb{N}_0$ let V_j be the space of functions on $[0,1)$ that are affine on every interval of $\mathcal{D}_j$; thus $\dim V_j = 2^{j+1}$ and $V_0 \subset V_1 \subset \cdots$. Put $D_0 := V_0$ and $D_j := V_j \ominus V_{j-1}$ (the orthogonal complement of V_{j-1} in V_j) for $j \ge 1$, so that $\dim D_j = 2^j$ for $j \ge 1$ and the spaces D_j are mutually orthogonal in $L^2[0,1)$. For $\ell \in \mathbb{N}_0^d$ let $D_\ell := D_{\ell_1} \otimes \cdots \otimes D_{\ell_d} \subset L^2([0,1)^d)$, and define

$$H := h + 3d, \qquad \mathcal{V}_H := \bigoplus_{\ell\in\mathbb{N}_0^d,\ |\ell|_1\le H} D_\ell. \tag{5.1}$$

Every element of $\mathcal{V}_H$ has a unique representative that is multiaffine, that is, affine in each coordinate separately, on each of the 2^{dH} half-open dyadic boxes with sides in $\mathcal{D}_H$; we always use this representative, which is defined at every point of $[0,1)^d$.

For a finite-dimensional space X of pointwise defined functions on a set Ω with a probability measure μ, the *inverse Christoffel function* is $L_X(p) := \sum_k |\phi_k(p)|^2$, where (ϕ_k) is an orthonormal basis of X in $L^2(\mu)$. It does not depend on the choice of the basis, since $L_X(p)$ is the squared norm of the evaluation functional $f \mapsto f(p)$ on X. It is nondecreasing in X, additive for orthogonal direct sums, and multiplicative for tensor products of spaces on product measure spaces. Theorem C requires a uniform bound on the inverse Christoffel function of the sampling space; in the proof of Lemma 5.3, the lower bound in part (c) of the following lemma keeps the weights bounded by C_d/n after a change of density, while part (b) controls the number of points.

Lemma 5.1 (Inverse Christoffel function of $\mathcal{V}_H$)**.** (a) *There are absolute constants $c_0, C_0 > 0$ such that $c_0\, 2^j \le L_{D_j}(t) \le C_0\, 2^j$ for all $j \in \mathbb{N}_0$ and $t \in [0,1)$.*

(b) $c_d\, 2^H (H+1)^{d-1} \le \dim \mathcal{V}_H \le C_d\, 2^H (H+1)^{d-1}$*; in particular* $c_d M \le \dim \mathcal{V}_H \le C_d M$.

(c) $c_d \dim \mathcal{V}_H \le L_{\mathcal{V}_H}(p) \le C_d \dim \mathcal{V}_H$ *for all* $p \in [0,1)^d$.

Proof. (a) For $I = [x_I, x_I + 2^{-j}) \in \mathcal{D}_j$, the functions $2^{j/2}\mathbf{1}_I$ and $2^{j/2}\sqrt{3}\,(2^{j+1}(t-x_I)-1)\mathbf{1}_I$ form an orthonormal basis of the functions that are affine on I and vanish outside I, so for $t \in I$,

$$L_{V_j}(t) = 2^j\big(1 + 12(s-\tfrac12)^2\big), \qquad s := 2^j(t-x_I) \in [0,1).$$

For $j = 0$ this gives $1 \le L_{D_0} = L_{V_0} \le 4$. For $j \ge 1$ and t in the left half I of $I' \in \mathcal{D}_{j-1}$ we have $2^{j-1}(t - x_{I'}) = s/2$, hence

$$L_{D_j}(t) = L_{V_j}(t) - L_{V_{j-1}}(t) = 2^{j-1}\big(4 + 21s^2 - 18s\big) \in \big[\tfrac17\, 2^{j-1},\, 7\cdot 2^{j-1}\big],$$

and the case of the right half is symmetric.

(b) Since $2^{|\ell|_1} \le \dim D_\ell = \prod_k \dim D_{\ell_k} \le 2^d 2^{|\ell|_1}$, and since $\sum_{|\ell|_1 \le H} 2^{|\ell|_1} = \sum_{b=0}^{H} 2^b \binom{b+d-1}{d-1}$ lies between $2^H\binom{H+d-1}{d-1}$ and $2^{H+1}\binom{H+d-1}{d-1}$, we get $\dim \mathcal{V}_H \asymp 2^H (H+1)^{d-1}$. By (5.1), (3.2) and (3.3), this is $\asymp (h+1)^{d-1}/v \asymp M$.

(c) By additivity and multiplicativity, $L_{\mathcal{V}_H} = \sum_{|\ell|_1 \le H} \prod_k L_{D_{\ell_k}}$, and (a) gives $c_0^d \sum_{|\ell|_1 \le H} 2^{|\ell|_1} \le L_{\mathcal{V}_H} \le C_0^d \sum_{|\ell|_1 \le H} 2^{|\ell|_1}$, which is $\asymp \dim \mathcal{V}_H$ by the proof of (b). □

Let $\mathcal{L} := J^*(\mathcal{H}_{\mathrm{lo}})$ be the space of primitives of the low Haar cross (4.1); since J^* is injective on $L^2([0,1]^d)$, $\dim \mathcal{L} = \dim \mathcal{H}_{\mathrm{lo}}$.

Lemma 5.2 (Space inclusions)**.** (a) $T_R \in \mathcal{V}_H$ *for every* $R \in \mathcal{R}$.

(b) $\mathcal{L} \subset \mathcal{V}_H$.

(c) $T_R \perp \mathcal{L}$ *for every* $R \in \mathcal{R}$.

Proof. (a) By Lemma 2.2, ψ_{R_k} is affine on every interval of $\mathcal{D}_{r_k+3}$, so $\psi_{R_k} \in V_{r_k+3} = \bigoplus_{\ell \le r_k+3} D_\ell$ and $T_R \in \bigoplus_{\ell_k \le r_k+3\ \forall k} D_\ell \subset \mathcal{V}_H$, because $\sum_k (r_k+3) = h + 3d = H$.

(b) Let $\varphi = \varphi_1 \otimes \cdots \otimes \varphi_d$ be a tensor Haar function of level at most $h-1$. By (2.4), $J^*\mathbf{1} = 1 - t \in V_0$ and $J^* h_J = -\tau_J \in V_{\mathrm{depth}(J)+1}$, so $J^*\varphi_k \in V_{\mathrm{lev}(\varphi_k)+1}$ and $J^*\varphi \in \bigoplus_{\ell_k \le \mathrm{lev}(\varphi_k)+1\ \forall k} D_\ell$, where $\sum_k (\mathrm{lev}(\varphi_k)+1) \le h - 1 + d \le H$. Since $\mathcal{L}$ is spanned by these $J^*\varphi$, (b) follows.

(c) With φ as in (b), $\langle T_R, J^*\varphi\rangle = v^{-1}\prod_k \langle \psi_{R_k}, J^*\varphi_k\rangle$. If some $\varphi_k = \mathbf{1}$, then $J^*\varphi_k = 1 - t$ is affine on $[0,1)$ and the k-th factor vanishes by Lemma 2.2. Otherwise $\varphi_k = h_{J_k}$ for every k, and $\sum_k \operatorname{depth}(J_k) \le h - 1 < h = \sum_k r_k$, so $\operatorname{depth}(J_k) \le r_k - 1$ for some k. Then τ_{J_k} is affine on each half of J_k, which are intervals of depth at most r_k, and R_k is either disjoint from J_k or contained in one of its halves; in both cases $\langle \psi_{R_k}, \tau_{J_k}\rangle = 0$ by Lemma 2.2, and (c) follows by linearity. □

5.2 Sampling with one additional function

For $u \in [-1,1]^{\mathcal{R}}$ let

$$\Delta_u := J^* P_{\mathrm{hi}} W(u) - F_{\mathrm{lin}}(u), \qquad w(p) := \mathbb{E}_U\big(|\Delta_U(p)|^2\big), \qquad p \in [0,1]^d, \tag{5.2}$$

with U uniform on $[-1,1]^{\mathcal{R}}$. By (4.5) and the orthogonality of the monomials $m_\Gamma(U)$,

$$w(p) = \sum_{\Gamma\in\mathcal{M},\ |\Gamma|\ge 3} c_\Gamma^2\, 3^{-|\Gamma|}\, \big|(J^* P_{\mathrm{hi}} X_\Gamma)(p)\big|^2$$

is a finite sum; each $P_{\mathrm{hi}} X_\Gamma = X_\Gamma - P_{\mathrm{lo}} X_\Gamma$ is bounded, so $J^* P_{\mathrm{hi}} X_\Gamma$ is Lipschitz continuous, and therefore w is continuous on $[0,1]^d$. Integrating this identity over the cube and using (4.6) with $\mathcal{O} = J^* P_{\mathrm{hi}}$ and Theorem 4.1 (with $\lambda \le \theta$) gives

$$\int_{[0,1]^d} w(p)\, dp = \mathbb{E}_U\big(\|\Delta_U\|_2^2\big) \le C_d\, \lambda^2 (h+1)^{d-1} v^2\, \theta^4. \tag{5.3}$$

To transfer this mean-square remainder bound to the sample points, we add the single function $\sqrt{w}$ to the sampling space.

Let $\Omega \subset (0,1)^d$ be the set of points of $[0,1)^d$ none of whose coordinates is an integer multiple of 2^{-H}. It has full measure and is the union of the open dyadic boxes with sides of depth H; every function in $\mathcal{V}_H$ is continuous on each of these open boxes, which is used to make the sample points distinct.

Lemma 5.3 (Sampling lemma)**.** *Let $w\colon [0,1]^d \to [0,\infty)$ be continuous. There are n distinct points $P_1, \dots, P_n \in \Omega$ and weights $a_1, \dots, a_n > 0$ such that*

$$c_d M \le n \le C_d M, \qquad a_i \le C_d/n, \qquad i \le n, \tag{5.4}$$

and, with absolute constants $c_\star, C_\star > 0$,

$$c_\star \|f\|_2^2 \le \sum_{i=1}^n a_i |f(P_i)|^2 \le C_\star \|f\|_2^2 \qquad \text{for all } f \in \mathcal{V}_H, \tag{5.5}$$

$$\sum_{i=1}^n a_i\, w(P_i) \le C_\star \int_{[0,1]^d} w(p)\, dp. \tag{5.6}$$

Proof. We use the change-of-density argument from the proof of [DPS+21, Theorem 6.4] (Theorem 6.3 in arXiv:2001.10636v1); see also [LT22, Remark 3.1]. We include the details to obtain distinct points and the required bound on the weights. Let $\mathcal{U}$ be the span of $\mathcal{V}_H$ and $\sqrt{w}$, considered as functions on Ω. All functions in $\mathcal{U}$ are continuous on each open dyadic box of depth H, so a function in $\mathcal{U}$ that vanishes almost everywhere vanishes on Ω, and $L^2(\Omega)$ with Lebesgue measure induces an inner product on $\mathcal{U}$; note that $\dim \mathcal{U} \in \{\dim \mathcal{V}_H, \dim \mathcal{V}_H + 1\}$. Let $(\phi_k)_{k \le \dim \mathcal{U}}$ be an orthonormal basis of $\mathcal{U}$ in $L^2(\Omega)$, and put $\varpi := L_{\mathcal{U}}/\dim \mathcal{U} = (\dim \mathcal{U})^{-1} \sum_k |\phi_k|^2$. Then $\int_\Omega \varpi = 1$, and since L_X is

nondecreasing in X, Lemma 5.1(c) gives $\varpi \geq L_{\mathcal{V}_H}/\dim \mathcal{U} \geq c_d \dim \mathcal{V}_H/(\dim \mathcal{V}_H + 1) \geq c'_d > 0$ on Ω. The functions $u_k := \phi_k/\sqrt{\varpi}$ are orthonormal in $L^2(\Omega, \varpi\, dp)$ and satisfy $\sum_k |u_k|^2 = \dim \mathcal{U}$ on Ω. By Theorem C, applied with $\mu = \varpi\, dp$, there are points $\xi^1, \dots, \xi^m \in \Omega$ with $\dim \mathcal{U} \leq m \leq C_1 \dim \mathcal{U}$ such that for every $g = f/\sqrt{\varpi}$ with $f \in \mathcal{U}$,

$$C_2 \|f\|_2^2 = C_2 \|g\|_{L^2(\mu)}^2 \leq \frac{1}{m} \sum_{j=1}^{m} \frac{|f(\xi^j)|^2}{\varpi(\xi^j)} \leq C_3 \|g\|_{L^2(\mu)}^2 = C_3 \|f\|_2^2.$$

With $\tilde{a}_j := 1/(m\varpi(\xi^j)) \leq 1/(mc'_d)$ this is a two-sided inequality $C_2\|f\|_2^2 \leq \sum_j \tilde{a}_j |f(\xi^j)|^2 \leq C_3\|f\|_2^2$ for all $f \in \mathcal{U}$, which contains (5.5) and, for $f = \sqrt{w}$, (5.6).

The points ξ^j need not be distinct. Let $\delta_0 > 0$, to be chosen below. Like the functions in $\mathcal{U}$, the functions ϖ and $1/(m\varpi)$ are continuous on each open dyadic box of depth H. If several points coincide, we replace them by points in the same open box at distance less than δ_0 from the original ones, chosen so that all m points become distinct, and recompute the weights $\tilde{a}_j = 1/(m\varpi(\xi^j)) \leq 1/(mc'_d)$. The functional $f \mapsto \sum_j \tilde{a}_j |f(\xi^j)|^2$ depends continuously on the points, and the unit sphere of $\mathcal{U}$ is compact, so for δ_0 small enough the two-sided inequality persists with the constants $C_2/2$ and $2C_3$. We take $n := m$, $P_i := \xi^i$ and $a_i := \tilde{a}_i$; then (5.5) and (5.6) hold with $c_\star := C_2/2$ and $C_\star := 2C_3$, and n lies between $\dim \mathcal{U} \geq \dim \mathcal{V}_H \geq c_d M$ and $C_1(\dim \mathcal{V}_H + 1) \leq C_d M$ by Lemma 5.1(b), so that $a_i \leq C_d/n$. □

The point set. From now on, $P_1, \dots, P_n \in \Omega$ and $a_1, \dots, a_n$ denote the points and weights provided by Lemma 5.3 for the function w in (5.2). They are deterministic: they depend on h and d only, and neither on z nor on any realization of U. Section 7 applies the hereditary-discrepancy argument to this point set.

5.3 The Gaussian weights and the response

We use the following matrices: the weight matrix $E := \operatorname{diag}(\sqrt{a_1}, \dots, \sqrt{a_n}) \in \mathbb{R}^{n \times n}$; the evaluation matrix $T := \big(T_R(P_i)\big)_{i \leq n,\, R \in \mathcal{R}} \in \mathbb{R}^{n \times M}$ of the primitives; the evaluation matrix $\Lambda \in \mathbb{R}^{n \times \dim \mathcal{L}}$ at the points of a fixed orthonormal basis of $\mathcal{L}$, so that the vectors Λb, $b \in \mathbb{R}^{\dim \mathcal{L}}$, are exactly the vectors $(g(P_i))_{i \leq n}$ with $g \in \mathcal{L}$; and the orthogonal projection $\Pi \in \mathbb{R}^{n \times n}$ onto the orthogonal complement of range$(E\Lambda)$. The *Gram matrix* is

$$\mathcal{G} := T^* E \Pi E T \in \mathbb{R}^{M \times M}. \tag{5.7}$$

Lemma 5.4 (Gram matrix)**.** *Let $c_\star, C_\star$ be the constants of (5.5). Then $c_\star \gamma_d v\, I_M \preceq \mathcal{G} \preceq C_\star \gamma_d v\, I_M$.*

Proof. For $y \in \mathbb{R}^M$ put $F_y := \sum_R y_R T_R$, so that $ETy = (\sqrt{a_i} F_y(P_i))_i$. Since Π projects onto range$(E\Lambda)^\perp$,

$$y^* \mathcal{G} y = \|\Pi E T y\|_2^2 = \min_b \|ETy - E\Lambda b\|_2^2 = \min_{g \in \mathcal{L}} \sum_i a_i \big|F_y(P_i) - g(P_i)\big|^2.$$

By Lemma 5.2, $F_y - g \in \mathcal{V}_H$ and $F_y \perp g$ for every $g \in \mathcal{L}$, so (5.5) and Lemma 3.1 give $\sum_i a_i |F_y(P_i) - g(P_i)|^2 \geq c_\star \|F_y - g\|_2^2 \geq c_\star \|F_y\|_2^2 = c_\star \gamma_d v \|y\|_2^2$ for every g, and hence $y^* \mathcal{G} y \geq c_\star \gamma_d v \|y\|_2^2$. Omitting the projection, $y^* \mathcal{G} y \leq \|ETy\|_2^2 = \sum_i a_i |F_y(P_i)|^2 \leq C_\star \|F_y\|_2^2 = C_\star \gamma_d v \|y\|_2^2$. □

We now define the scale α and the matrix B by

$$\alpha := \omega_d \sqrt{n/v}, \qquad B := \alpha\gamma_d v \, E\Pi E T \, \mathcal{G}^{-1} \in \mathbb{R}^{n\times M}, \tag{5.8}$$

where $\omega_d > 0$ is a constant depending only on d, chosen in the next lemma. The two factors of B serve different purposes. The projection Π removes the evaluated primitives of the low Haar cross: the weight vectors Bz are orthogonal to every vector $(g(P_i))_{i\le n}$ with $g \in \mathcal{L}$ (Lemma 5.5(a)), so that the weighted counting function does not see the low Haar cross. The inverse Gram matrix makes the pairing with the primitives T_R exactly diagonal, $T^*B = \alpha\gamma_d v \, I_M$ (Lemma 5.5(b)); hence the linear part of the response pairs each coordinate z_R only with u_R (Proposition 5.6(d)), which matches Section 6, where the correlations are controlled coordinate by coordinate (Proposition 6.2).

Lemma 5.5 (Properties of B)**.** (a) $\Lambda^* B = 0$.

(b) $T^*B = \alpha\gamma_d v \, I_M$.

(c) *If ω_d is sufficiently small, depending only on d, then $BB^* \preceq I_n$.*

Proof. (a) Since Π is symmetric and annihilates range$(E\Lambda)$, we have $\Lambda^* E\Pi = (\Pi E\Lambda)^* = 0$, and hence $\Lambda^* B = 0$. (b) $T^*B = \alpha\gamma_d v \, T^* E\Pi E T \, \mathcal{G}^{-1} = \alpha\gamma_d v \, \mathcal{G}\mathcal{G}^{-1}$. (c) We have $\|B\|_{\mathrm{op}} \le \alpha\gamma_d v \, \|E\|_{\mathrm{op}} \|ET\|_{\mathrm{op}} \|\mathcal{G}^{-1}\|_{\mathrm{op}}$, with $\|E\|_{\mathrm{op}}^2 = \max_i a_i \le C_d/n$, $\|ET\|_{\mathrm{op}}^2 = \|T^* E^2 T\|_{\mathrm{op}} \le C_\star \gamma_d v$ by (5.5) and Lemma 3.1, and $\|\mathcal{G}^{-1}\|_{\mathrm{op}} \le (c_\star \gamma_d v)^{-1}$ by Lemma 5.4. Hence $\|B\|_{\mathrm{op}}^2 \le C_d \, \alpha^2 \gamma_d v/n = C_d \gamma_d \omega_d^2$, which is at most one for small ω_d; this is equivalent to $BB^* \preceq I_n$. □

We fix ω_d as in Lemma 5.5(c). For $z \in \mathbb{R}^M$, the *weight vector* is $\eta := Bz \in \mathbb{R}^n$, and the associated *weighted counting function* is the function G_η of (1.3),

$$G_z(x) := \sum_{i=1}^{n} (Bz)_i \, \mathbf{1}_{\{P_i \le x\}}, \qquad x \in [0,1]^d. \tag{5.9}$$

For $u \in [-1,1]^{\mathcal{R}}$, the *response* of the point set to the test function $W(u)$ is

$$V(u) := B^* \big((J^* W(u))(P_i) \big)_{i\le n} \in \mathbb{R}^M. \tag{5.10}$$

The linear part of the response is exactly diagonal; the nonlinear part contributes the error $e(u)$, estimated in part (e) of the following proposition.

Proposition 5.6 (Exact linear response)**.** *Let $s_R := \alpha\gamma_d v^2 \lambda_R$ for $R \in \mathcal{R}$, $s := \alpha\gamma_d v^2 \lambda$, and $s_* := \min_R s_R$, so that $s/2 \le s_* \le s_R \le s$ by Lemma 3.3. Then the following hold.*

(a) $\langle G_z, W(u)\rangle = z \cdot V(u)$ *for all* $z \in \mathbb{R}^M$ *and* $u \in [-1,1]^{\mathcal{R}}$.

(b) $P_{\mathrm{lo}} G_z = 0$ *for every* z, *and* $B^* \big((J^* f)(P_i) \big)_i = B^* \big((J^* P_{\mathrm{hi}} f)(P_i) \big)_i$ *for every* $f \in L^2([0,1]^d)$.

(c) $\langle G_z, q_R \rangle = \alpha\gamma_d v^2 \, z_R$ *for all* $z \in \mathbb{R}^M$ *and* $R \in \mathcal{R}$.

(d) $V(u) = \mathrm{diag}(s_R)\, u + e(u)$, *where* $e(u) := B^* \big(\Delta_u(P_i) \big)_{i\le n}$ *with* Δ_u *as in* (5.2).

(e) *If U is uniform on $[-1,1]^{\mathcal{R}}$, then $\mathbb{E}_U(\|e(U)\|_2^2) \le \rho_h^2 \, s_*^2 M$, where $\rho_h := C_\rho (h+1)^{-(d-2)}$ with a constant C_ρ depending only on d.*

(f) $c_d (h+1)^{d-1/2} \le s_* M \le sM \le C_d (h+1)^{d-1/2}$.

Proof. (a) By Fubini's theorem and (2.3),

$$\langle G_z, W(u)\rangle = \sum_i (Bz)_i \int_{[0,1]^d} \mathbf{1}_{\{P_i \le x\}} W(u;x)\,dx = \sum_i (Bz)_i\,(J^*W(u))(P_i) = z\cdot V(u).$$

(b) For $f \in L^2([0,1]^d)$, $J^*P_{\mathrm{lo}}f \in \mathcal{L}$, so $\big((J^*P_{\mathrm{lo}}f)(P_i)\big)_i = \Lambda b$ for some $b \in \mathbb{R}^{\dim \mathcal{L}}$, and $B^*\Lambda b = (\Lambda^* B)^* b = 0$ by Lemma 5.5(a). Since $J^*f = J^*P_{\mathrm{lo}}f + J^*P_{\mathrm{hi}}f$, this proves the second claim, and applying it to $f \in \mathcal{H}_{\mathrm{lo}}$ (where $P_{\mathrm{hi}}f = 0$) together with the computation in (a) gives $\langle G_z, f\rangle = 0$, that is, $P_{\mathrm{lo}}G_z = 0$.

(c) As in (a), $\langle G_z, q_R\rangle = \sum_i (Bz)_i\,(J^*q_R)(P_i) = v\sum_i (Bz)_i T_R(P_i) = v\,(T^*Bz)_R = \alpha\gamma_d v^2 z_R$ by Lemma 5.5(b).

(d) By (b), $V(u) = B^*\big((J^*P_{\mathrm{hi}}W(u))(P_i)\big)_i = B^*\big(F_{\mathrm{lin}}(u)(P_i)\big)_i + e(u)$. By (3.14), $\big(F_{\mathrm{lin}}(u)(P_i)\big)_i = vT\,\mathrm{diag}(\lambda_R)u$, and $B^*\big(vT\,\mathrm{diag}(\lambda_R)u\big) = v(T^*B)^*\,\mathrm{diag}(\lambda_R)u = \alpha\gamma_d v^2\,\mathrm{diag}(\lambda_R)u$ by Lemma 5.5(b).

(e) By (5.8), we have $e(u) = \alpha\gamma_d v\,\mathcal{G}^{-1}T^*E\Pi\,E\big(\Delta_u(P_i)\big)_i$. Since $\Pi^2 = \Pi = \Pi^*$, we have $\|T^*E\Pi\|_{\mathrm{op}}^2 = \|T^*E\Pi ET\|_{\mathrm{op}} = \|\mathcal{G}\|_{\mathrm{op}} \le C_\star\gamma_d v$, and $\|\mathcal{G}^{-1}\|_{\mathrm{op}} \le (c_\star\gamma_d v)^{-1}$ by Lemma 5.4, so that

$$\|e(u)\|_2^2 \le \alpha^2\gamma_d^2 v^2 \cdot \frac{C_\star\gamma_d v}{c_\star^2\gamma_d^2 v^2}\sum_i a_i|\Delta_u(P_i)|^2 = \frac{C_\star}{c_\star^2}\,\alpha^2\gamma_d v\sum_i a_i|\Delta_u(P_i)|^2.$$

Taking expectations and using (5.6) and (5.3),

$$\mathbb{E}_U\big(\|e(U)\|_2^2\big) \le \frac{C_\star}{c_\star^2}\,\alpha^2\gamma_d v\sum_i a_i w(P_i) \le \frac{C_\star^2}{c_\star^2}\,\alpha^2\gamma_d v\int_{[0,1]^d} w \le C_d\,\alpha^2 v^3\lambda^2(h+1)^{d-1}\theta^4.$$

On the other hand, $s^2M = \alpha^2\gamma_d^2 v^4\lambda^2 K_h/v \ge c_d\alpha^2 v^3\lambda^2(h+1)^{d-1}$ by (3.2). Hence $\mathbb{E}_U(\|e(U)\|_2^2) \le C_d\theta^4 s^2 M \le 4C_d\theta^4 s_*^2 M = \rho_h^2 s_*^2 M$ with $C_\rho := 2C_d^{1/2}t_d^2$, where C_d denotes the constant in the preceding inequality and we used $\theta^4 = t_d^4(h+1)^{-2(d-2)}$; since α^2 has cancelled, C_ρ does not depend on ω_d.

(f) By (5.8), (3.3) and (5.4), $sM = \omega_d\sqrt{n/v}\,\gamma_d v\lambda K_h$ with $n \asymp M = K_h/v$, so $sM \asymp \sqrt{K_h}\,v^{-1}\cdot v\lambda K_h = \lambda K_h^{3/2} \asymp \lambda(h+1)^{3(d-1)/2}$ by (3.2). Inserting $\lambda = \varepsilon t_d(h+1)^{-(d-2)/2}$ gives $sM \asymp (h+1)^{d-1/2}$, and $s_* \ge s/2$. □

6 The entropy obstruction

We now study the standard Gaussian vector $z \in \mathbb{R}^M$ conditioned on the event that all weighted box counts are small. For $t > 0$ let

$$K_t := \big\{z \in \mathbb{R}^M : \|G_z\|_\infty \le t\big\}, \qquad \|G_z\|_\infty := \sup_{x\in[0,1]^d}|G_z(x)|, \tag{6.1}$$

with G_z as in (5.9). With the enumeration of $\mathcal{R}$ fixed in Section 3, $\mathbb{R}^M = \mathbb{R}^{\mathcal{R}}$ carries the coordinates $z_1, \dots, z_M$.

Lemma 6.1 (Geometry of K_t)**.** *For every $t > 0$, the set K_t is closed, convex, symmetric and bounded, and it contains a neighborhood of the origin. In particular $0 < \gamma_M(K_t)$.*

Proof. For fixed x, the map $z \mapsto G_z(x)$ is linear, and $G_z(x)$ takes only finitely many values as x ranges over the cube, namely the values $\sum_{i\in S}(Bz)_i$ over the finitely many incidence patterns $S = \{i : P_i \le x\}$. Hence $z \mapsto \|G_z\|_\infty$ is a seminorm on $\mathbb{R}^M$, and K_t is closed, convex and symmetric; being the unit ball of a seminorm on a finite-dimensional space, it contains a neighborhood of the origin. By Proposition 5.6(c), $\alpha\gamma_d v^2|z_R| = |\langle G_z, q_R\rangle| \le \|G_z\|_\infty \|q_R\|_1 = v\|G_z\|_\infty$, so $K_t \subseteq [-t/(\alpha\gamma_d v), t/(\alpha\gamma_d v)]^M$. □

6.1 Conditioned Gaussian vectors and Rosenblatt coordinates

Let $K \subset \mathbb{R}^M$ be a bounded Borel set with $p := \gamma_M(K) > 0$, and let $\nu := \gamma_M(\cdot \mid K)$ be the conditioned Gaussian measure. It has the density

$$\varphi(z) = \frac{(2\pi)^{-M/2}\mathrm{e}^{-\|z\|_2^2/2}}{p}\,\mathbf{1}_K(z), \tag{6.2}$$

which is bounded and vanishes outside the bounded set K. Hence Lemma 2.6 applies to a random vector Z with law ν, and

$$\mathsf{h}(\nu) = -\int \varphi\log\varphi = \frac{M}{2}\log(2\pi) + \frac{1}{2}\,\mathbb{E}_\nu\big(\|Z\|_2^2\big) + \log p \;\ge\; \frac{M}{2}\log(2\pi) + \log p. \tag{6.3}$$

We only use the lower bound in (6.3): since $\mathsf{h}(\gamma_M) = \frac{M}{2}\log(2\pi\mathrm{e})$, the differential entropy of the conditioned vector is at most $\log(1/p) + M/2$ below that of γ_M, that is, at most $\kappa + \frac{1}{2}$ nats per coordinate when $p \ge \mathrm{e}^{-\kappa M}$. Let $U = U(Z)$ be the Rosenblatt transform of Z under ν, as in Lemma 2.5: $U_j := 2F_j(Z_j \mid Z_{<j}) - 1$, where $F_j(\cdot \mid z_{<j})$ is the conditional distribution function of Z_j given $Z_{<j} = z_{<j}$ under ν.

Proposition 6.2 (Correlation with the Rosenblatt coordinates)**.** *Let K, p, ν, Z and U be as above. Then U is uniformly distributed on $[-1,1]^M$ under ν, $\mathbb{E}_\nu(Z_jU_j) \ge 0$ for every $j \le M$, and*

$$\sum_{j=1}^{M}\mathbb{E}_\nu(Z_jU_j) \;\ge\; \frac{M}{4\mathrm{e}}\exp\Big(\frac{\mathsf{h}(\nu)}{M}\Big) \;\ge\; \frac{\sqrt{2\pi}}{4\mathrm{e}}\,p^{1/M}\,M. \tag{6.4}$$

In particular, if $p \ge \mathrm{e}^{-\kappa M}$ for some $\kappa > 0$, then $\sum_j \mathbb{E}_\nu(Z_jU_j) \ge c_\kappa M$ with $c_\kappa := \frac{\sqrt{2\pi}}{4\mathrm{e}}\mathrm{e}^{-\kappa}$.

Proof. The uniform law of U is Lemma 2.5. We first prove a scalar inequality. Let Y be a real random variable with a bounded density of bounded support, let F be its distribution function, and let Y' be an independent copy. Since F is continuous, $\mathbb{E}(\operatorname{sgn}(Y - Y') \mid Y) = F(Y) - (1 - F(Y)) = 2F(Y) - 1$ almost surely, and exchanging the roles of Y and Y',

$$\mathbb{E}(Y\,(2F(Y)-1)) = \mathbb{E}\big(Y\operatorname{sgn}(Y-Y')\big) = \tfrac{1}{2}\,\mathbb{E}\big((Y-Y')\operatorname{sgn}(Y-Y')\big) = \tfrac{1}{2}\,\mathbb{E}\big(|Y-Y'|\big) \ge 0. \tag{6.5}$$

Conditioning on Y and using Jensen's inequality, $\mathbb{E}(|Y - Y'|) \ge \mathbb{E}(|Y - \mathbb{E}(Y')|) = \mathbb{E}(|Y - \mathbb{E}\,Y|)$, and Lemma 2.7 gives $\mathbb{E}(|Y - \mathbb{E}\,Y|) \ge \mathrm{e}^{\mathsf{h}(Y)}/(2\mathrm{e})$. Hence

$$\mathbb{E}(Y\,(2F(Y)-1)) \;\ge\; \frac{\mathrm{e}^{\mathsf{h}(Y)}}{4\mathrm{e}}. \tag{6.6}$$

Now fix j. For $\nu_{<j}$-almost every $z_{<j}$, where $\nu_{<j}$ is the law of $Z_{<j}$ (namely for those $z_{<j}$ at which the marginal density of $Z_{<j}$ is positive), the conditional law of Z_j given $Z_{<j} = z_{<j}$ has a bounded

density of bounded support, namely $\varphi_{\le j}(z_{<j},\cdot)/\varphi_{<j}(z_{<j})$ in terms of the marginal densities of φ. Applying (6.5) and (6.6) to this conditional law gives

$$\mathbb{E}_\nu(Z_jU_j \mid Z_{<j}=z_{<j}) \;\ge\; \frac{1}{4\mathrm{e}}\exp\big(\mathsf{h}(Z_j \mid Z_{<j}=z_{<j})\big) \;\ge\; 0.$$

Taking the expectation over $z_{<j}$ and applying Jensen's inequality to the convex function exp yields $\mathbb{E}_\nu(Z_jU_j) \ge \frac{1}{4\mathrm{e}}\exp\big(\mathsf{h}(Z_j \mid Z_{<j})\big)$, with the averaged conditional entropy of Lemma 2.6. Summing over j, applying Jensen's inequality once more, and using the chain rule,

$$\sum_{j=1}^{M}\mathbb{E}_\nu(Z_jU_j) \;\ge\; \frac{M}{4\mathrm{e}}\exp\Big(\frac{1}{M}\sum_{j=1}^{M}\mathsf{h}(Z_j \mid Z_{<j})\Big) = \frac{M}{4\mathrm{e}}\exp\Big(\frac{\mathsf{h}(\nu)}{M}\Big).$$

The second inequality in (6.4) follows from (6.3). □

6.2 The small-ball obstruction

We combine Proposition 6.2 with the exact linear response of Proposition 5.6. Recall that $s_* = \min_R s_R$ and that $\rho_h = C_\rho(h+1)^{-(d-2)}$ is the quantity in Proposition 5.6(e).

Proposition 6.3 (Small-ball obstruction)**.** *Let $\kappa > 0$ and $t > 0$ be such that $\gamma_M(K_t) \ge \mathrm{e}^{-\kappa M}$. Then*

$$t \;\ge\; (c_\kappa - \rho_h)\, s_* M, \qquad c_\kappa = \frac{\sqrt{2\pi}}{4\mathrm{e}}\mathrm{e}^{-\kappa}.$$

Proof. Let $\nu := \gamma_M(\,\cdot \mid K_t)$, let Z have law ν, and let $U = U(Z)$ be its Rosenblatt transform; by Lemma 6.1, K_t satisfies the assumptions of Section 6.1. For every $z \in K_t$ and every $u \in [-1,1]^{\mathcal{R}}$, Proposition 5.6(a) and Lemma 3.2 give

$$z \cdot V(u) = \langle G_z, W(u)\rangle \le \|G_z\|_\infty \, \|W(u)\|_1 \le t.$$

Since $Z \in K_t$ almost surely and $U \in [-1,1]^{\mathcal{R}}$, it follows that $t \ge \mathbb{E}_\nu(Z \cdot V(U))$; the expectation exists because Z is bounded on K_t and V is a polynomial on a compact set. By Proposition 5.6(d),

$$\mathbb{E}_\nu(Z \cdot V(U)) = \sum_{R\in\mathcal{R}} s_R\, \mathbb{E}_\nu(Z_RU_R) + \mathbb{E}_\nu(Z \cdot e(U)).$$

Each term $\mathbb{E}_\nu(Z_RU_R)$ is nonnegative by Proposition 6.2, so the first sum is at least

$$s_* \sum_{R\in\mathcal{R}} \mathbb{E}_\nu(Z_RU_R) \;\ge\; s_* c_\kappa M.$$

For the second term, the Cauchy–Schwarz inequality, Lemma 2.4 and the fact that U has exactly the uniform law on $[-1,1]^{\mathcal{R}}$ under ν give

$$\big|\,\mathbb{E}_\nu(Z \cdot e(U))\big| \le \big(\mathbb{E}_\nu\big(\|Z\|_2^2\big)\big)^{1/2}\big(\mathbb{E}_\nu\big(\|e(U)\|_2^2\big)\big)^{1/2} \le \sqrt{M}\cdot\big(\mathbb{E}_U\big(\|e(U)\|_2^2\big)\big)^{1/2} \le \sqrt{M}\cdot\rho_h s_*\sqrt{M},$$

by Proposition 5.6(e). Hence $t \ge s_*c_\kappa M - \rho_h s_* M$. Note that no independence between Z and $e(U(Z))$ is used, and that the Rosenblatt transform is never differentiated. □

7 Proof of the main theorem

It remains to show that the hereditary discrepancy D of the point set gives $\gamma_M(K_{2D}) \geq \mathrm{e}^{-\kappa M}$ with κ depending only on d (Step 1 of the overview), and to pass from the constructed point sets to all sufficiently large N.

Lemma 7.1 (Gaussian small balls from hereditary discrepancy)**.** *Let $A \in \mathbb{R}^{m\times n}$ satisfy* $\operatorname{herdisc}(A) \leq D$*, and let η be a centered Gaussian vector in $\mathbb{R}^n$ with* $\operatorname{var}(\eta_i) \leq 1$ *for all $i \leq n$. Then*

$$\mathbb{P}(\|A\eta\|_\infty \leq 2D) \geq (p_0/2)^n, \qquad p_0 := \mathbb{P}(|g| \leq 1),$$

where g is a standard Gaussian variable.

The argument is a Gaussian analogue of the volume argument behind the determinant lower bound of Lovász, Spencer and Vesztergombi [LSV86, Lemma 2 and Theorem 2].

Proof. By Theorem A, $\mathbb{P}(\eta \in [-1,1]^n) \geq \prod_i \mathbb{P}(|\eta_i| \leq 1) \geq p_0^n$, because $\mathbb{P}(|\eta_i| \leq 1) \geq p_0$ for a centered Gaussian variable of variance at most one. By (2.2), the event $\{\eta \in [-1,1]^n\}$ is contained in the union over $\sigma \in \{-1,1\}^n$ of the events $\{A\eta \in A\sigma + K\}$ with $K := [-2D, 2D]^m$. The vector $A\eta$ is a centered Gaussian vector in $\mathbb{R}^m$, and K is closed, convex and symmetric, so Theorem B gives $\mathbb{P}(A\eta \in A\sigma + K) \leq \mathbb{P}(A\eta \in K)$ for every σ. By the union bound over the 2^n vectors σ, $p_0^n \leq \mathbb{P}(\eta \in [-1,1]^n) \leq 2^n\, \mathbb{P}(A\eta \in K)$. □

Let $h \in 3\mathbb{N}$, let $P = \{P_1, \dots, P_n\} \subset (0,1)^d$ be the point set of Section 5 at depth h, and let A_P be the incidence matrix of the set system $\{P \cap A(x) : x \in [0,1]^d\}$, as in Section 2.1. Put

$$\mathfrak{D}_h := \operatorname{herdisc}(A_P). \tag{7.1}$$

Proposition 7.2 (Hereditary discrepancy of the construction)**.** *There are constants $c_d > 0$ and $h_d \in 3\mathbb{N}$ such that the point set of Section 5 at depth h satisfies $\mathfrak{D}_h \geq c_d(h+1)^{d-1/2}$ for all $h \in 3\mathbb{N}$ with $h \geq h_d$.*

Proof. Let z be standard Gaussian in $\mathbb{R}^M$ and $\eta := Bz$. By Lemma 5.5(c), $\operatorname{var}(\eta_i) = (BB^*)_{ii} \leq 1$. The rows of A_P are the vectors $(\mathbf{1}_{\{P_i \leq x\}})_{i\leq n}$, $x \in [0,1]^d$, so $\|A_P\eta\|_\infty = \sup_x |G_z(x)| = \|G_z\|_\infty$ and $\{\|A_P\eta\|_\infty \leq 2\mathfrak{D}_h\} = \{z \in K_{2\mathfrak{D}_h}\}$. Note that $\mathfrak{D}_h \geq 1$, so that $t = 2\mathfrak{D}_h > 0$ is admissible in Proposition 6.3: the restriction of A_P to a single column has discrepancy one, because the point P_i lies in $A(x)$ for $x = (1, \dots, 1)$. Lemma 7.1 with $D = \mathfrak{D}_h$ and (5.4) give

$$\gamma_M(K_{2\mathfrak{D}_h}) \geq (p_0/2)^n = \mathrm{e}^{-a_0 n} \geq \mathrm{e}^{-\kappa_d M}, \qquad a_0 := \log(2/p_0), \quad \kappa_d := a_0 C_d,$$

where C_d is the constant with $n \leq C_d M$ in (5.4). Proposition 6.3 with $t = 2\mathfrak{D}_h$ and $\kappa = \kappa_d$ yields $2\mathfrak{D}_h \geq (c_{\kappa_d} - \rho_h)s_* M$. Here $c_{\kappa_d} > 0$ depends only on d, while $\rho_h = C_\rho(h+1)^{-(d-2)}$ decreases to zero as $h \to \infty$ because $d \geq 3$; choose $h_d \in 3\mathbb{N}$ such that $\rho_h \leq c_{\kappa_d}/2$ for all $h \geq h_d$. For $h \geq h_d$ we obtain $2\mathfrak{D}_h \geq \frac{1}{2} c_{\kappa_d} s_* M$, and Proposition 5.6(f) gives $\mathfrak{D}_h \geq \frac{1}{4} c_{\kappa_d} s_* M \geq c_d(h+1)^{d-1/2}$. □

Proof of Theorem 1.1. By (5.4), (3.3) and (3.2), the construction at depth h has $n \leq C'_d(h+1)^{d-1}2^h$ points, with a constant $C'_d \geq 1$. For sufficiently large N, let $h = h(N)$ be the largest element of $3\mathbb{N}$ with $C'_d(h+1)^{d-1}2^h \leq N$. Then $h \leq \log_2 N$, and the maximality of h gives $C'_d(h+4)^{d-1}2^{h+3} > N$, hence $h > \log_2 N - \log_2(8C'_d(\log_2 N + 4)^{d-1}) \geq \frac{1}{2}\log_2 N$ for all $N \geq N_d$, where N_d is chosen so large

that for every $N \geq N_d$ the number $h(N)$ is defined, $8C'_d(\log_2 N + 4)^{d-1} \leq \sqrt{N}$ and $\log_2 N \geq 2h_d$; in particular $h(N) \geq h_d$.

For such N, the point set P at depth $h(N)$ has $n \leq N$ points, and Proposition 7.2 together with (2.1) provides a subset $P' \subseteq P$ such that every coloring χ of P' satisfies $\sup_{x\in[0,1]^d} |\sum_{p\in P'\cap A(x)} \chi(p)| \geq \mathfrak{D}_h \geq c_d(h+1)^{d-1/2} \geq c'_d(\log N)^{d-1/2}$ for a constant $c'_d > 0$ depending only on d.

Let P'' be the union of P' with $N - |P'|$ distinct points all of whose coordinates lie in $(1, 2)$. For every $x \in [0,1]^d$ we have $P'' \cap A(x) = P' \cap A(x)$, since the added points have no coordinate at most one, and every coloring of P'' restricts to a coloring of P'; as the supremum defining $\mathrm{disc}(P'', \mathcal{A}_d)$ runs over all $x \in \mathbb{R}^d$, in particular over $x \in [0,1]^d$, we get $\mathrm{disc}(P'', \mathcal{A}_d) \geq c'_d(\log N)^{d-1/2}$. Since $\mathrm{disc}(N, \mathcal{A}_d)$ is the supremum over all N-point sets in $\mathbb{R}^d$, this proves Theorem 1.1 with the constant c'_d. Moreover, dilating P'' by $1/2$ gives an N-point set in $(0,1)^d$ satisfying the same lower bound. □

Remark 7.3 (Order of the constants)**.** For fixed d, the constants are chosen in the following order, none of them depending on h: $\varepsilon = \frac{1}{4}$ and $t_d = t^*_d$ (Theorem 4.1); the sampling constants (Lemma 5.3), which together with t_d fix C_ρ and hence the sequence $(\rho_h)_h$; ω_d (Lemma 5.5), which fixes α, hence s and s_* for every h, and the constants in Proposition 5.6(f); the entropy constant $\kappa_d = a_0 C_d$ and hence c_{κ_d}; finally h_d and c_d (Proposition 7.2), and then N_d and c'_d (Theorem 1.1). Although κ_d depends on the sampling constant C_d through $n \leq C_d M$, there is no circularity: the sampling constants are not adapted to κ_d, because ρ_h decays polynomially in h for every fixed $d \geq 3$, while c_{κ_d} is a fixed positive number.

8 Extensions and algorithmic consequences

This section proves Corollary 1.3 and gives an efficient coloring algorithm attaining the upper bound in Corollary 1.2. The algorithm is independent of Theorem 1.1; that theorem establishes the optimality of its guarantee for $d \geq 3$.

8.1 Fixed convex polytopes

Recall that $\mathcal{T}_B = \{tB + x : t > 0,\ x \in \mathbb{R}^d\}$ denotes the family of homothetic copies of a convex polytope $B \subset \mathbb{R}^d$. For a finite set $P \subset \mathbb{R}^d$ and a set $F \subseteq \mathbb{R}^d$ we call $P \cap F$ the *trace* of F on P.

Lemma 8.1 (Finite-trace transfer)**.** *Let $d \geq 1$, let $B \subset \mathbb{R}^d$ be a convex polytope with nonempty interior and let $P \subset \mathbb{R}^d$ be finite. There are an invertible linear map L of $\mathbb{R}^d$ and $t > 0$ such that for every $x \in \mathbb{R}^d$ there is $y \in \mathbb{R}^d$ with $L(P \cap A(x)) = L(P) \cap (tB + y)$.*

Proof. The idea is to find a linear map under which, for the finitely many relevant differences $x - p$, membership in the tangent cone of B at a vertex coincides with $p \leq x$; choosing t large then makes the facets of B away from this vertex irrelevant.

Translating B does not change $\mathcal{T}_B$, so we may assume that the origin is a vertex of B. Write

$$B = \{y \in \mathbb{R}^d : \ell_i(y) \geq 0 \text{ for } i \leq m,\ a_r(y) \leq b_r \text{ for } r \leq m'\},$$

where the linear functionals $\ell_1, \dots, \ell_m$ define the facets of B through the origin, and the linear functionals $a_1, \dots, a_{m'}$ with constants $b_1, \dots, b_{m'}$ define the remaining facets; $b_r > 0$ because the origin belongs to B but lies on none of the latter facets. The tangent cone $C := \{y : \ell_i(y) \geq 0 \text{ for } i \leq m\}$ has nonempty interior because it contains B, and it is pointed: since all $b_r > 0$, a vector $y \neq 0$

with $\ell_i(y) = 0$ for all i would give a segment $\{sy : |s| \le s_0\}$ in B lying in all facets through the vertex 0, whose intersection is $\{0\}$. Moreover, $tB = \{y \in C : a_r(y) \le tb_r \text{ for } r \le m'\}$ for $t > 0$.

A basis adapted to C. Choose faces $\{0\} = F_0 \subset F_1 \subset \cdots \subset F_d = C$ of C with $\dim F_j = j$, which exist because C is pointed and every proper face of a polyhedral cone is contained in a face of the next dimension, and choose v_j in the relative interior of F_j for $j = 1, \dots, d$. Then $v_1, \dots, v_d$ is a basis of $\mathbb{R}^d$. Indeed, the face $F_{j-1} = F_j \cap H$ is cut out of F_j by a supporting linear hyperplane H of F_j (it passes through the origin since F_j is a cone), so that $\operatorname{span} F_{j-1} \subseteq H$; the relative interior point v_j of F_j lies in H only if $F_j \subseteq H$, which is excluded by $\dim F_j > \dim F_{j-1}$. Hence $v_j \notin \operatorname{span}\{v_1, \dots, v_{j-1}\}$. For $\varepsilon > 0$ let T_ε be the invertible linear map $T_\varepsilon z := \sum_{j=1}^{d} \varepsilon^{j-1} z_j v_j$. Since $v_j \in C$, all $\ell_i(v_j)$ are nonnegative, and since v_d is an interior point of C, all $\ell_i(v_d)$ are positive. Put $k(i) := \min\{j : \ell_i(v_j) > 0\}$. Then $\ell_i(v_j) = 0$ for $j < k(i)$, so that for every fixed $z \in \mathbb{R}^d$

$$\varepsilon^{1-k(i)} \ell_i(T_\varepsilon z) = \sum_{j \ge k(i)} \varepsilon^{j-k(i)} z_j \ell_i(v_j) \longrightarrow \ell_i(v_{k(i)})\, z_{k(i)}, \qquad \varepsilon \downarrow 0. \tag{8.1}$$

Moreover, every $j \in \{1, \dots, d\}$ is of the form $k(i)$: the face F_{j-1} is the intersection of the facets of C containing it, and $F_j \not\subseteq F_{j-1}$, so some facet $C \cap \{\ell_i = 0\}$ contains F_{j-1} but not F_j; then $\ell_i(v_{j'}) = 0$ for $j' < j$, while $\ell_i \ge 0$ on F_j and $\ell_i \not\equiv 0$ on F_j force $\ell_i(v_j) > 0$ at the relative interior point v_j, that is, $k(i) = j$.

Traces. Let $X \subset \mathbb{R}^d$ be a finite set containing, for every trace $P \cap A(x)$ with $x \in \mathbb{R}^d$, one point x with this trace, chosen such that $x_j \ne p_j$ for all $p \in P$ and $j \le d$; this is possible because increasing a coordinate of x slightly does not change the trace. All coordinates of the finitely many differences $x - p$ with $x \in X$ and $p \in P$ are nonzero, so by (8.1) there is $\varepsilon > 0$ such that $\ell_i(T_\varepsilon(x - p))$ has the sign of $(x - p)_{k(i)}$ for all i, x and p. As the indices $k(i)$ exhaust $\{1, \dots, d\}$, we obtain

$$T_\varepsilon(x - p) \in C \iff (x - p)_j > 0 \text{ for all } j \le d \iff p \le x, \qquad p \in P,\ x \in X,$$

where the last equivalence uses $x_j \ne p_j$. Fix this ε, put $L := -T_\varepsilon$, and choose $t > 0$ with $a_r(L(p - x)) \le tb_r$ for all $p \in P$, $x \in X$ and $r \le m'$. Then, for $p \in P$ and $x \in X$,

$$L(p) \in tB + L(x) \iff L(p - x) \in C \iff T_\varepsilon(x - p) \in C \iff p \le x,$$

where the first equivalence uses the choice of t; that is, $L(P) \cap (tB + L(x)) = L(P \cap A(x))$. Since every trace $P \cap A(x)$ with $x \in \mathbb{R}^d$ equals $P \cap A(x')$ for some $x' \in X$, the claim follows with $y := L(x')$. □

Proof of Corollary 1.3. Let $P \subset \mathbb{R}^d$ be any N-point set and let L and t be as in Lemma 8.1. The set $Q := L(P)$ has N points, the colorings of Q are the maps $\chi \circ L^{-1}$ with χ a coloring of P, and by Lemma 8.1, L maps every set $P \cap A(x)$ onto a set $Q \cap (tB + y)$ with $tB + y \in \mathcal{T}_B$. Hence $\operatorname{disc}(Q, \mathcal{T}_B) \ge \operatorname{disc}(P, \mathcal{A}_d)$, and taking the supremum over P gives $\operatorname{disc}(N, \mathcal{T}_B) \ge \operatorname{disc}(N, \mathcal{A}_d) \ge c_d (\log N)^{d-1/2}$ for $N \ge N_d$ by Theorem 1.1. The upper bound $\operatorname{disc}(N, \mathcal{T}_B) = O_{d,B}((\log N)^{d-1/2})$ is [Nik17, Theorem 2]. □

For generic polytopes, [Nik17, Theorem 3] gives the lower bound $\Omega_{d,B}((\log N)^{d-1})$; Nikolov conjectured that the genericity assumption there is unnecessary and that the lower-bound constant can be chosen independently of B [Nik17, p. 1093]. Both conjectures already follow from Lemma 8.1: the proof of Corollary 1.3 gives $\operatorname{disc}(N, \mathcal{T}_B) \ge \operatorname{disc}(N, \mathcal{A}_d)$ in every dimension, and $\operatorname{disc}(N, \mathcal{A}_d) = \Omega_d((\log N)^{d-1})$ for $d \ge 2$ [MN15, MNT20]. The bound for all sufficiently large N follows from the grid estimate in [MNT20, Section 6] by taking a subset realizing hereditary discrepancy and padding, as in the proof of Theorem 1.1. For $d \ge 3$, Corollary 1.3 improves the exponent to $d - \frac{1}{2}$, with the constants c_d, N_d of Theorem 1.1.

8.2 A matching offline coloring guarantee

We use the following recent result on prefix discrepancy. For operation counts, we count arithmetic operations and comparisons on real numbers, evaluations of square roots and exponentials, and sampling from $\mathcal{N}(0,1)$ or Uniform$[0,1)$ at unit cost.

Theorem D (Aden-Ali [AA26, Theorem 1]). *Let $v_1, \dots, v_T \in \mathbb{R}^m$ be a fixed sequence with $\|v_i\|_2 \le 1$ for all $i \le T$; in particular, the vectors do not depend on the random choices of the algorithm below. There is a randomized algorithm which receives $v_1, \dots, v_T$ one at a time and chooses a sign $\sigma_i \in \{-1, 1\}$ upon receiving v_i, before seeing $v_{i+1}, \dots, v_T$, with the following property: for every $k \le T$, the prefix sum $S_k := \sum_{i \le k} \sigma_i v_i$ can be written as $S_k = G_{k,1} + G_{k,2} + G_{k,3}$ with random vectors $G_{k,1}, G_{k,2}, G_{k,3}$ each having the standard Gaussian distribution on $\mathbb{R}^m$ (they need not be independent). In the model above, the algorithm runs in time $O(mT)$.*

By Hölder's inequality, $\mathbb{E}(\exp(\langle S_k, y\rangle)) \le \prod_{j=1}^{3} \big(\mathbb{E}(\exp(3\langle G_{k,j}, y\rangle))\big)^{1/3} = \exp(9\|y\|_2^2/2)$ for every $y \in \mathbb{R}^m$, so every prefix sum is subgaussian with variance proxy 9, as noted in [AA26, Section 1], and the Chernoff bound gives, for $w \in \mathbb{R}^m \setminus \{0\}$ and $u > 0$,

$$\mathbb{P}(|\langle S_k, w\rangle| \ge u) \le 2\exp\Big(-\frac{u^2}{18\|w\|_2^2}\Big). \tag{8.2}$$

We follow the reduction in Nikolov's proof of [Nik17, Theorem 1], using an explicit dyadic factorization and replacing Banaszczyk's signed-series theorem [Ban12] ([Nik17, Lemma 9]) with Theorem D.

Proposition 8.2 (Efficient offline coloring for anchored boxes). *Let $d \ge 2$. There is a randomized offline algorithm (it reads the whole point set before choosing the first sign) which, given an N-point set $P \subset \mathbb{R}^d$ with $N \ge 2$ and $\delta \in (0,1)$, outputs a coloring $\chi\colon P \to \{-1,1\}$ such that, with probability at least $1-\delta$,*

$$\sup_{x \in \mathbb{R}^d} \Big| \sum_{p \in P \cap A(x)} \chi(p) \Big| \le 3K \sqrt{2 \log \frac{2(N+1)^d}{\delta}}, \qquad K := (\lceil \log_2 N \rceil + 1)^{d-1}, \tag{8.3}$$

and on the same event the discrepancy of χ with respect to all axis-parallel boxes is at most 2^d times this bound. The algorithm performs $O_d(N^2 (\log N)^{d-1})$ operations in the model of Theorem D.

Proof. Reduction to ranks and prefixes. Put $H := \lceil \log_2 N \rceil$, so that $N \le 2^H$, and fix an order of the points of P. For $j \le d-1$ let $r_j(p) \in \{1, \dots, N\}$ be the rank of p among the points of P with respect to the j-th coordinate, ties broken by the fixed order, and let $r(p) := (r_1(p), \dots, r_{d-1}(p))$. For every threshold x_j, the points with $p_j \le x_j$ are exactly the points with $r_j(p) \le \rho_j$, where $\rho_j \in \{0, \dots, N\}$ is their number, because tied points are included together. Enumerate the points as $P_1, \dots, P_N$ in nondecreasing order of the last coordinate, ties again broken by the fixed order; then the points p with $p_d \le x_d$ are $P_1, \dots, P_k$ for some $k \in \{0, \dots, N\}$. Consequently every trace $P \cap A(x)$ has the form

$$\{P_i : i \le k,\ r(P_i) \le \rho\} \qquad \text{with } k \in \{0, \dots, N\} \text{ and } \rho \in \{0, \dots, N\}^{d-1}, \tag{8.4}$$

and there are at most $(N+1)^d$ distinct traces.

Dyadic products. For $0 \le l \le H$ the dyadic integer intervals $\{(q-1)2^l + 1, \dots, q2^l\}$, $1 \le q \le 2^{H-l}$, partition $\{1, \dots, 2^H\}$, so every rank vector $r(p)$ lies in exactly $K = (H+1)^{d-1}$ products of dyadic intervals, one for each choice of the levels $l_1, \dots, l_{d-1}$. Let $\mathcal{Q}$ be the set of dyadic products containing at least one rank vector $r(P_i)$; then $m := |\mathcal{Q}| \le NK$. For $i \le N$ let $v_i \in \mathbb{R}^{\mathcal{Q}}$ be the indicator vector

of the products in $\mathcal{Q}$ containing $r(P_i)$, divided by $\sqrt{K}$, so that $\|v_i\|_2 = 1$. The algorithm applies Theorem D to $v_1, \dots, v_N$ in this order and outputs $\chi(P_i) := \sigma_i$.

Discrepancy. Fix a trace (8.4). Each initial segment $\{1, \dots, \rho_j\}$ of $\{1, \dots, 2^H\}$ is the disjoint union of at most $H+1$ dyadic intervals, one for each binary digit of ρ_j, so $\{s \in \{1, \dots, 2^H\}^{d-1} : s \le \rho\}$ is the disjoint union of at most K dyadic products. Let $w \in \{0,1\}^{\mathcal{Q}}$ be the indicator vector of those among them that belong to $\mathcal{Q}$; then $\|w\|_2^2 \le K$. The traces and the vectors w depend only on P, not on the random signs. For every point P_i of the trace, $r(P_i)$ lies in exactly one of these products, and for no other $i \le k$ does $r(P_i)$ lie in any of them, so

$$\sum_{i \le k,\ r(P_i) \le \rho} \sigma_i = \sqrt{K}\, \langle S_k, w \rangle, \qquad S_k = \sum_{i \le k} \sigma_i v_i.$$

If $k = 0$ or $w = 0$, the signed count vanishes. Otherwise (8.2) gives $\mathbb{P}(\sqrt{K}|\langle S_k, w\rangle| \ge u) \le 2\exp(-u^2/(18K^2))$ for $u > 0$. With $u := 3K\sqrt{2\log(2(N+1)^d/\delta)}$ this probability is at most $\delta(N+1)^{-d}$, and a union bound over the at most $(N+1)^d$ traces gives (8.3) with probability at least $1-\delta$. For an axis-parallel box in $\mathcal{B}_d$, the indicator function of its trace on P is a signed combination of the indicator functions of 2^d traces $P \cap A(x)$ by inclusion–exclusion, as in (1.1), which gives the factor 2^d on the same event.

Operations. Sorting and ranking take $O(dN \log N)$ comparisons. The NK incidences between points and dyadic products can be enumerated and indexed in $O_d(NK \log(NK))$ operations. Constructing the vectors $v_1, \dots, v_N$ and applying Theorem D takes $O(mN) = O(N^2K) = O_d(N^2(\log N)^{d-1})$ operations, since $m \le NK$, and this dominates. Since the ranks and the order of the points depend on the whole input, the algorithm is offline. □

For $\delta = N^{-a}$ with fixed $a > 0$, the bound (8.3) is $O_{d,a}((\log N)^{d-1/2})$, since $K \asymp (\log N)^{d-1}$ and $\log(2(N+1)^d/\delta) = O_{d,a}(\log N)$. For $d \ge 3$, Theorem 1.1 shows that no coloring algorithm can guarantee a smaller order of worst-case discrepancy, so the discrepancy guarantee of Proposition 8.2 is optimal up to the constant. The operation count refers to the model of Theorem D; finite-precision (bit) complexity is not addressed.

Acknowledgments

The author used Anthropic's Claude and OpenAI's Codex to assist with literature research, proof development and checking, and manuscript preparation. The author retains full responsibility for the content of the manuscript, including all mathematical claims and references.

References

[AA26] Ishaq Aden-Ali. Optimal online discrepancy minimization in linear time, 2026. arXiv:2607.04388v1.

[ABN18] Christoph Aistleitner, Dmitriy Bilyk, and Aleksandar Nikolov. Tusnády's problem, the transference principle, and non-uniform QMC sampling. In *Monte Carlo and quasi-Monte Carlo methods*, volume 241 of *Springer Proc. Math. Stat.*, pages 169–180. Springer, Cham, 2018.

[And55] T. W. Anderson. The integral of a symmetric unimodal function over a symmetric convex set and some probability inequalities. *Proc. Amer. Math. Soc.*, 6(2):170–176, 1955.

[Ban12] Wojciech Banaszczyk. On series of signed vectors and their rearrangements. *Random Structures Algorithms*, 40(3):301–316, 2012.

[Ban23] Nikhil Bansal. Discrepancy theory and related algorithms. In *Proceedings of the International Congress of Mathematicians 2022. Vol. 7*, pages 5178–5210. EMS Press, 2023.

[Bec81] József Beck. Balanced two-colorings of finite sets in the square. I. *Combinatorica*, 1(4):327–335, 1981.

[Bec89] József Beck. Balanced two-colorings of finite sets in the cube. *Discrete Math.*, 73(1-2):13–25, 1989.

[BG17] Nikhil Bansal and Shashwat Garg. Algorithmic discrepancy beyond partial coloring. In *STOC'17—Proceedings of the 49th Annual ACM SIGACT Symposium on Theory of Computing*, pages 914–926. ACM, New York, 2017.

[BL08] Dmitriy Bilyk and Michael T. Lacey. On the small ball inequality in three dimensions. *Duke Math. J.*, 143(1):81–115, 2008.

[BLV08] Dmitriy Bilyk, Michael T. Lacey, and Armen Vagharshakyan. On the small ball inequality in all dimensions. *J. Funct. Anal.*, 254(9):2470–2502, 2008.

[BN26] Nikhil Bansal and Aleksandar Nikolov. Discrepancy theory: an algorithmic and geometric perspective, 2026. arXiv:2608.00140.

[Bog98] Vladimir I. Bogachev. *Gaussian measures*, volume 62 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 1998.

[Boh90] Géza Bohus. On the discrepancy of 3 permutations. *Random Structures Algorithms*, 1(2):215–220, 1990.

[BS95] József Beck and Vera T. Sós. Discrepancy theory. In *Handbook of combinatorics, Vol. 1, 2*, pages 1405–1446. Elsevier Sci. B. V., Amsterdam, 1995.

[CT06] Thomas M. Cover and Joy A. Thomas. *Elements of information theory*. Wiley-Interscience [John Wiley & Sons], Hoboken, NJ, second edition, 2006.

[DPS+21] F. Dai, A. Prymak, A. Shadrin, V. Temlyakov, and S. Tikhonov. Entropy numbers and Marcinkiewicz-type discretization. *J. Funct. Anal.*, 281(6):Paper No. 109090, 25, 2021.

[Hal81] G. Halász. On Roth's method in the theory of irregularities of point distributions. In *Recent progress in analytic number theory, Vol. 2 (Durham, 1979)*, pages 79–94. Academic Press, London-New York, 1981.

[Lar14] Kasper Green Larsen. On range searching in the group model and combinatorial discrepancy. *SIAM J. Comput.*, 43(2):673–686, 2014.

[LSV86] L. Lovász, J. Spencer, and K. Vesztergombi. Discrepancy of set-systems and matrices. *European J. Combin.*, 7(2):151–160, 1986.

[LT22] I. Limonova and V. Temlyakov. On sampling discretization in L_2. *J. Math. Anal. Appl.*, 515(2):Paper No. 126457, 14, 2022.

[Mat99a] Jiří Matoušek. *Geometric discrepancy*, volume 18 of *Algorithms and Combinatorics*. Springer-Verlag, Berlin, 1999. An illustrated guide.

[Mat99b] Jiří Matoušek. On the discrepancy for boxes and polytopes. *Monatsh. Math.*, 127(4):325–336, 1999.

[MN15] Jiří Matoušek and Aleksandar Nikolov. Combinatorial discrepancy for boxes via the γ_2 norm. In *31st International Symposium on Computational Geometry (SoCG 2015)*, volume 34 of *LIPIcs*, pages 1–15. Schloss Dagstuhl. Leibniz-Zent. Inform., Wadern, 2015.

[MNT20] Jiří Matoušek, Aleksandar Nikolov, and Kunal Talwar. Factorization norms and hereditary discrepancy. *Int. Math. Res. Not. IMRN*, 2020(3):751–780, 2020.

[Nik17] Aleksandar Nikolov. Tighter bounds for the discrepancy of boxes and polytopes. *Mathematika*, 63(3):1091–1113, 2017.

[O'D14] Ryan O'Donnell. *Analysis of Boolean functions*. Cambridge University Press, New York, 2014.

[Ros52] Murray Rosenblatt. Remarks on a multivariate transformation. *Ann. Math. Statistics*, 23(3):470–472, 1952.

[Rot54] K. F. Roth. On irregularities of distribution. *Mathematika*, 1(2):73–79, 1954.

[Sch72] Wolfgang M. Schmidt. Irregularities of distribution. VII. *Acta Arith.*, 21:45–50, 1972.

[Sri97] Aravind Srinivasan. Improving the discrepancy bound for sparse matrices: better approximations for sparse lattice approximation problems. In *Proceedings of the Eighth Annual ACM-SIAM Symposium on Discrete Algorithms*, pages 692–701. ACM, New York, 1997.

[Tal94] Michel Talagrand. The small ball problem for the Brownian sheet. *Ann. Probab.*, 22(3):1331–1354, 1994.

[Tem95a] V. N. Temlyakov. An inequality for trigonometric polynomials and its application for estimating the entropy numbers. *J. Complexity*, 11(2):293–307, 1995.

[Tem95b] V. N. Temlyakov. Some inequalities for multivariate Haar polynomials. *East J. Approx.*, 1(1):61–72, 1995.

[Š67] Zbyněk Šidák. Rectangular confidence regions for the means of multivariate normal distributions. *J. Amer. Statist. Assoc.*, 62(318):626–633, 1967.

[Wei26] Zhewei Wei. Tight bounds for Tusnády's problem in the plane, 2026. arXiv:2610.08130.